

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	5
2	Data correlation and clustering	6
2.1	Clustermat and cluster list	6
2.2	Features selection	7
3	Model(s) analysis	10
3.1	Metadata	10
3.2	Explanation legend tables	11
3.3	Repartition of train and test	12
3.3.1	External Split: X_train vs X_test	12
3.3.2	Internal CV Folds	12
3.4	Model : Linear	14
3.4.1	Gridsearch	14
3.4.2	Scatter	16
3.4.3	Scatter residuals	17
3.4.4	Learning curve	17
3.5	Model : LGBM	18
3.5.1	Gridsearch	18
3.5.2	Scatter	25
3.5.3	Scatter residuals	26
3.5.4	Learning curve	26
3.6	Model : CatBoost	27
3.6.1	Gridsearch	27
3.6.2	Scatter	34
3.6.3	Scatter residuals	35
3.6.4	Learning curve	35
3.7	Sumup-table	36

4	Plot	36
4.1	Standalone plots	36
4.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	36
4.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	39
4.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	42
4.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	45
4.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	48
4.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	51
4.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	54
4.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	57
4.2	Compare plots	60
4.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	60
4.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	62
4.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	63
4.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	64
4.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	66
4.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	67
4.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	68
4.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	70

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7730
Nombre de lignes df clean	7730
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5411
Taille dataset test	2319
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_tempera-

ture_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

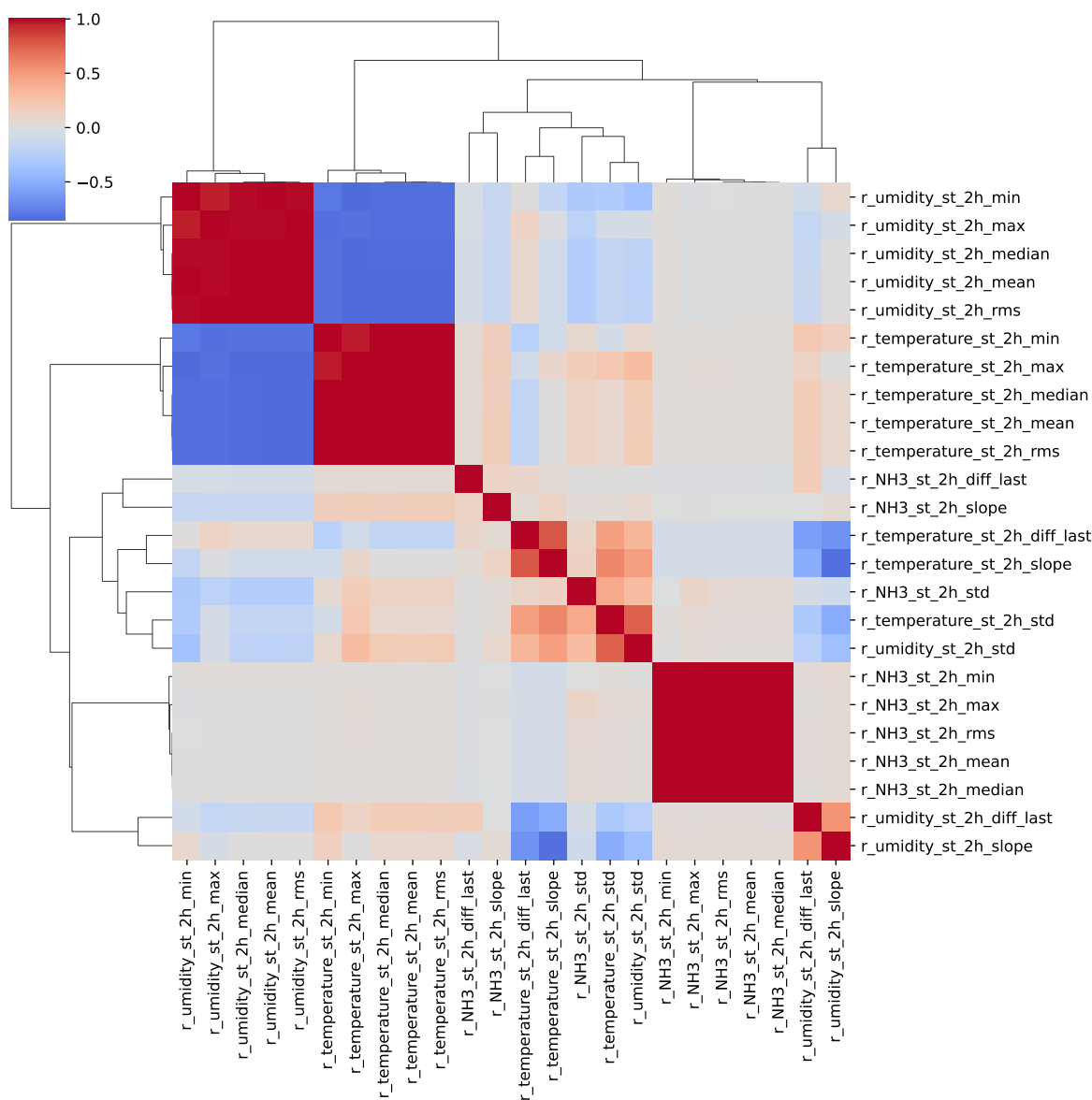
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_2h_min, r_NH3_st_2h_median, r_NH3_st_2h_mean, r_NH3_st_2h_rms, r_NH3_st_2h_max ; **cluster 7:** r_NH3_st_2h_std ; **cluster 8:** r_NH3_st_2h_diff_last ; **cluster 9:** r_NH3_st_2h_slope ; **cluster 10:** r_temperature_st_2h_mean, r_umidity_st_2h_mean, r_umidity_st_2h_min, r_umidity_st_2h_rms, r_umidity_st_2h_max, r_temperature_st_2h_max, r_temperature_st_2h_min, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_umidity_st_2h_median ; **cluster 11:** r_temperature_st_2h_std ; **cluster 12:** r_temperature_st_2h_diff_last ; **cluster 13:** r_umidity_st_2h_slope, r_temperature_st_2h_slope ; **cluster 14:** r_umidity_st_2h_std ; **cluster 15:** r_umidity_st_2h_diff_last ; **cluster 16:** r_NH3_st_8h_min, r_NH3_st_8h_rms, r_NH3_st_8h_max, r_NH3_st_8h_mean, r_NH3_st_8h_median ; **cluster 17:** r_NH3_st_8h_std ; **cluster 18:** r_NH3_st_8h_diff_last ; **cluster 19:** r_NH3_st_8h_slope ; **cluster 20:** r_umidity_st_8h_max, r_temperature_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_min, r_temperature_st_8h_mean, r_temperature_st_8h_median,

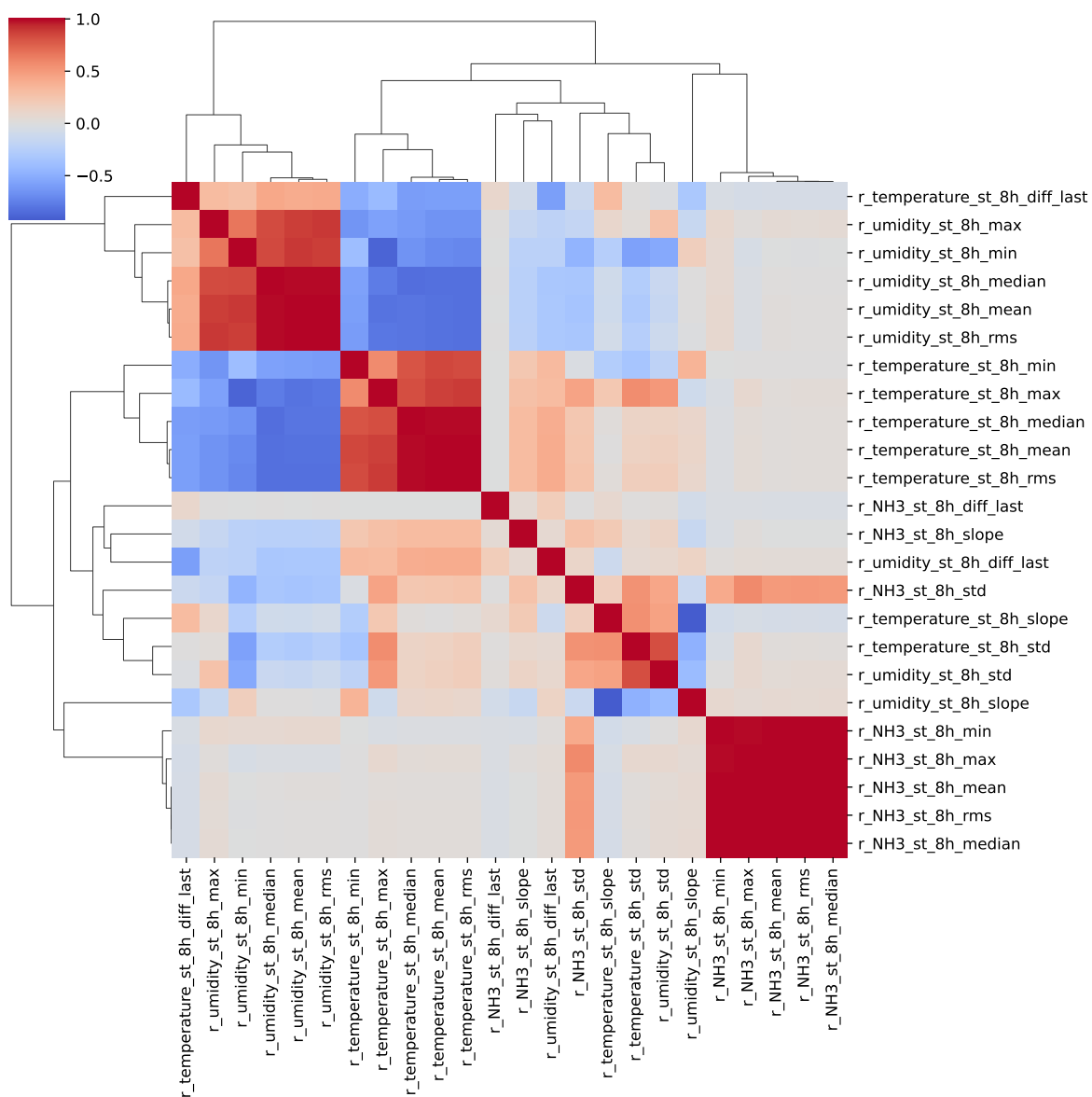
r_umidity_st_8h_mean, r_umidity_st_8h_rms ; **cluster 21:** r_umidity_st_8h_std, r_temperature_st_8h_std ; **cluster 22:** r_temperature_st_8h_diff_last ; **cluster 23:** r_temperature_st_8h_slope, r_umidity_st_8h_slope ; **cluster 24:** r_umidity_st_8h_diff_last ; **cluster 25:** r_NH3_st_12h_mean, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max ; **cluster 26:** r_NH3_st_12h_std ; **cluster 27:** r_NH3_st_12h_diff_last ; **cluster 28:** r_NH3_st_12h_slope ; **cluster 29:** r_umidity_st_12h_mean, r_temperature_st_12h_mean, r_umidity_st_12h_min, r_umidity_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_max, r_temperature_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_max ; **cluster 30:** r_temperature_st_12h_std ; **cluster 31:** r_temperature_st_12h_min ; **cluster 32:** r_temperature_st_12h_diff_last ; **cluster 33:** r_umidity_st_12h_slope, r_temperature_st_12h_slope ; **cluster 34:** r_umidity_st_12h_std ; **cluster 35:** r_umidity_st_12h_diff_last ;

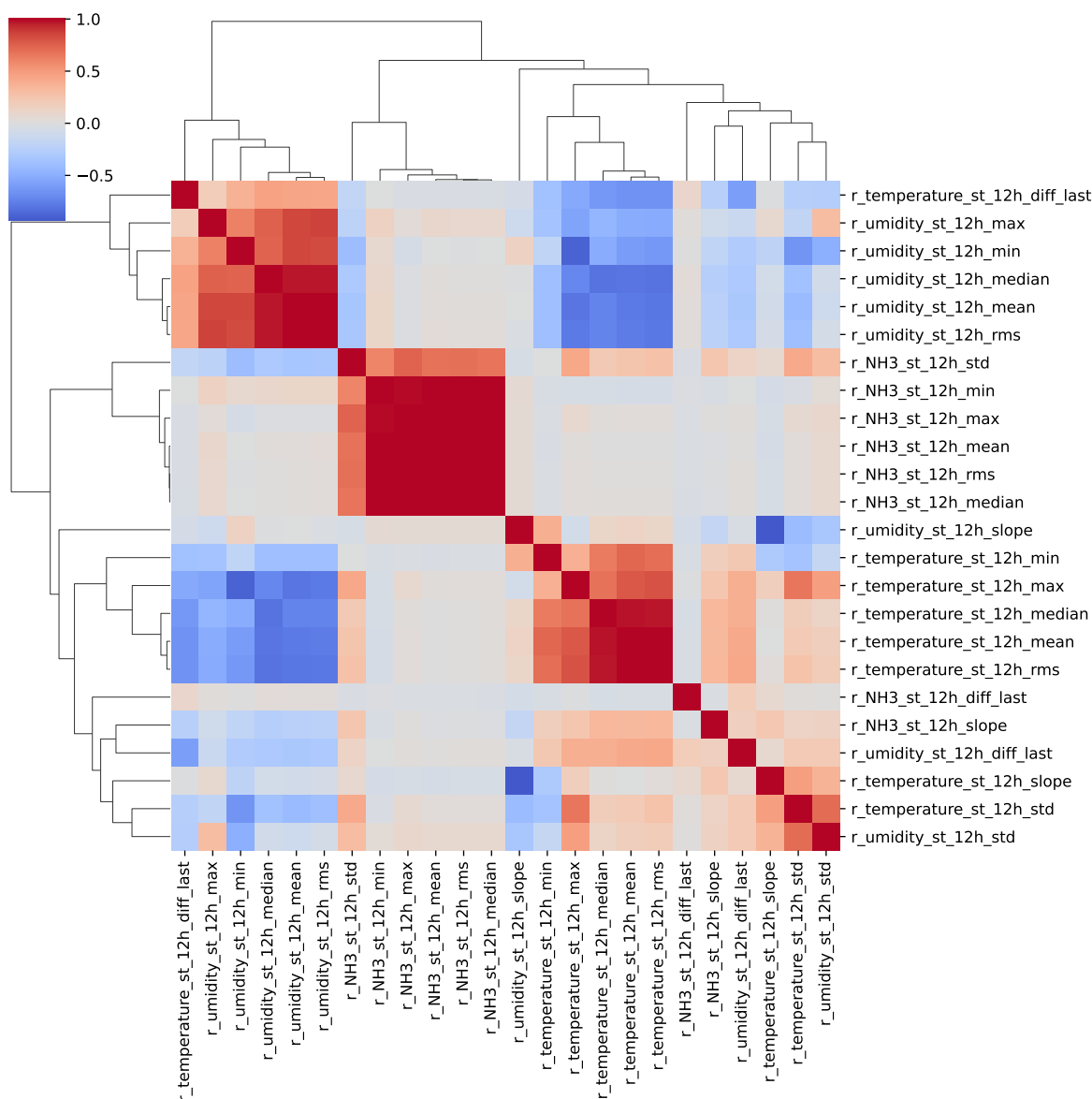
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features _X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_8h_min, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_max, r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_umidity_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last







3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umid-

ity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_8h_min, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_max, r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_umidity_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last

Len(Features_X): 35

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5411 samples Test: 2319 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



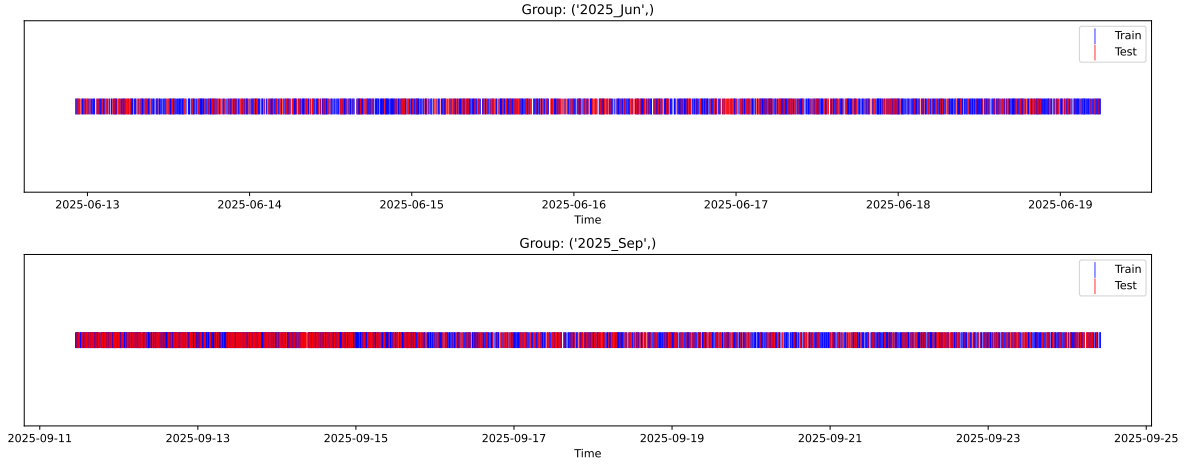
Fold 3/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



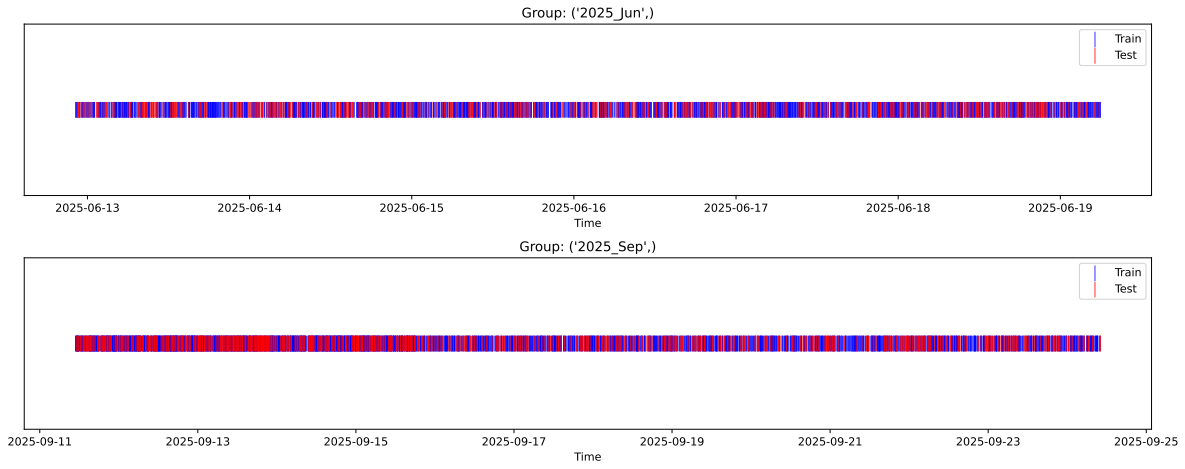
Fold 4/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.342, std=0.800, min=0.959, max=13.845
 y_{test} : mean=2.364, std=0.795, min=1.068, max=7.277

Fold 0 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.34, std=0.80

Fold 1 — y_{test} : mean=2.35, std=0.83, y_{train} : mean=2.34, std=0.79

Fold 2 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.81

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.85, y_train: mean=2.34, std=0.78

train scores CV: [0.56973837 0.59574981 0.57001608 0.56989782 0.58346733]

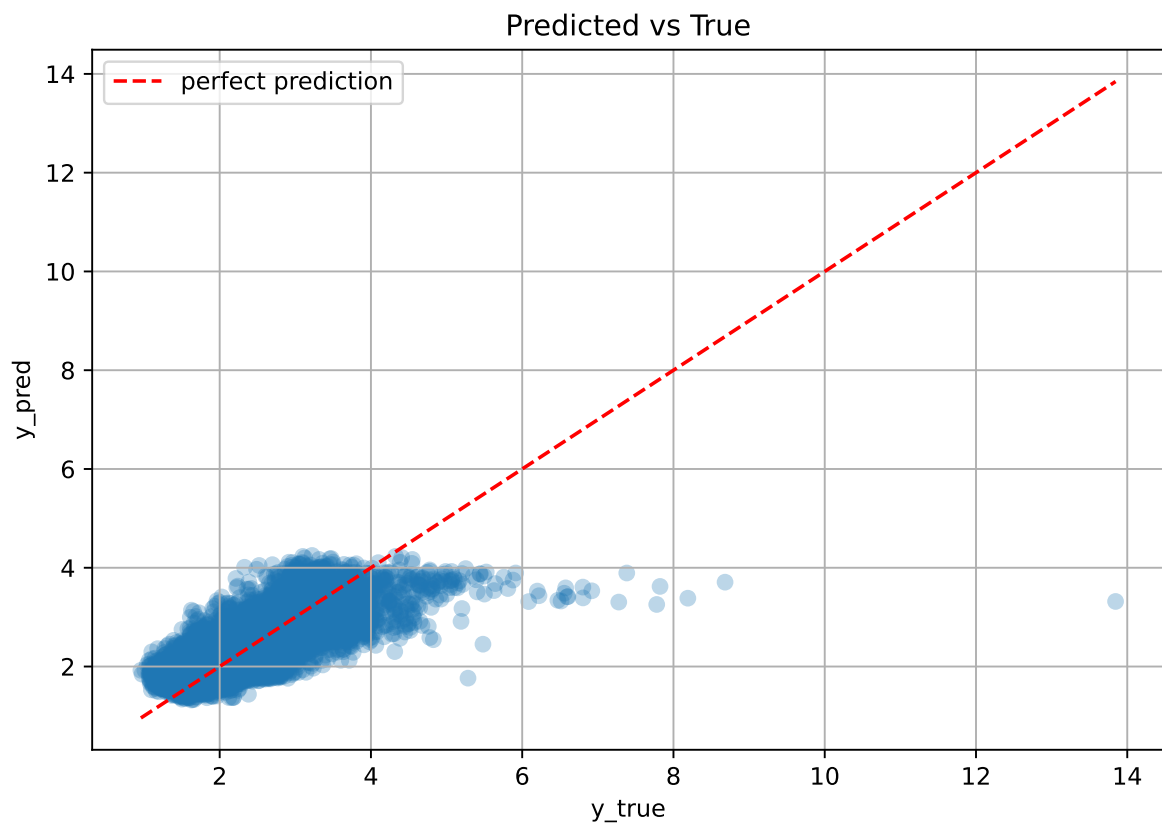
test scores CV: [0.57038552 0.5148495 0.56949482 0.57002446 0.54257983]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5778	0.5535	0.022	0.597	0.5986	0.0451	1.0816	1.0439

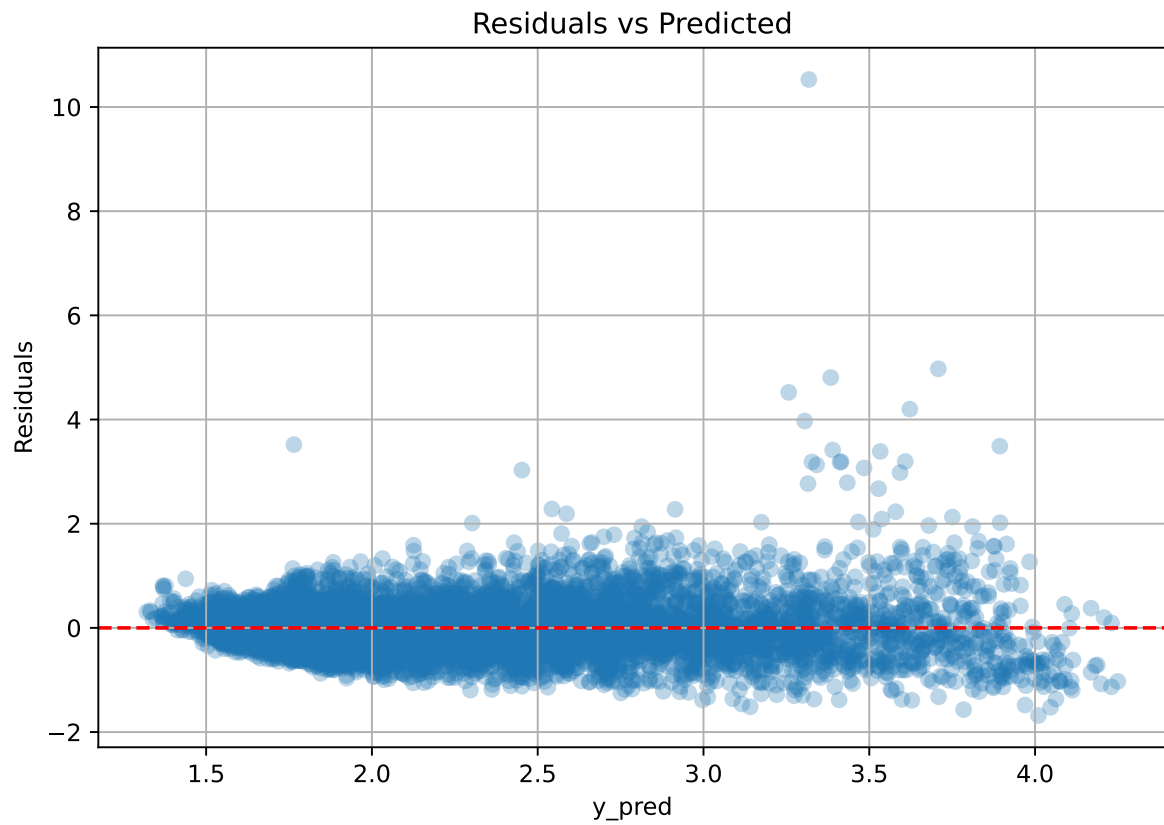
Coefficients:

coef_r_CO2 = -0.0432, coef_r_NH3 = 0.1659, coef_r_NH3_st_12h_diff_last = 0.0044, coef_r_NH3_st_12h_mean = 0.7785, coef_r_NH3_st_12h_slope = 0.0840, coef_r_NH3_st_12h_std = 0.0149, coef_r_NH3_st_2h_diff_last = 0.0044, coef_r_NH3_st_2h_min = -0.4087, coef_r_NH3_st_2h_slope = -0.0029, coef_r_NH3_st_2h_std = 0.0123, coef_r_NH3_st_8h_diff_last = 0.0044, coef_r_NH3_st_8h_min = -0.5817, coef_r_NH3_st_8h_slope = 0.0505, coef_r_NH3_st_8h_std = -0.0422, coef_r_THI = 0.4101, coef_r_temperature = 0.1771, coef_r_temperature_st_12h_diff_last = -0.0111, coef_r_temperature_st_12h_min = -0.0669, coef_r_temperature_st_12h_std = -0.0144, coef_r_temperature_st_2h_diff_last = -0.0111, coef_r_temperature_st_2h_mean = -0.4101, coef_r_temperature_st_2h_std = 0.0718, coef_r_temperature_st_8h_diff_last = -0.0111, coef_r_temperature_st_8h_slope = -0.0504, coef_r_umidity = -0.7122, coef_r_umidity_st_12h_diff_last = -0.0074, coef_r_umidity_st_12h_mean = 0.2292, coef_r_umidity_st_12h_slope = 0.3542, coef_r_umidity_st_12h_std = -0.0823, coef_r_umidity_st_2h_diff_last = -0.0074, coef_r_umidity_st_2h_slope = 0.0423, coef_r_umidity_st_2h_std = 0.0389, coef_r_umidity_st_8h_diff_last = -0.0074, coef_r_umidity_st_8h_max = 0.0095, coef_r_umidity_st_8h_std = -0.0430, intercept = 2.3417,

3.4.2 Scatter



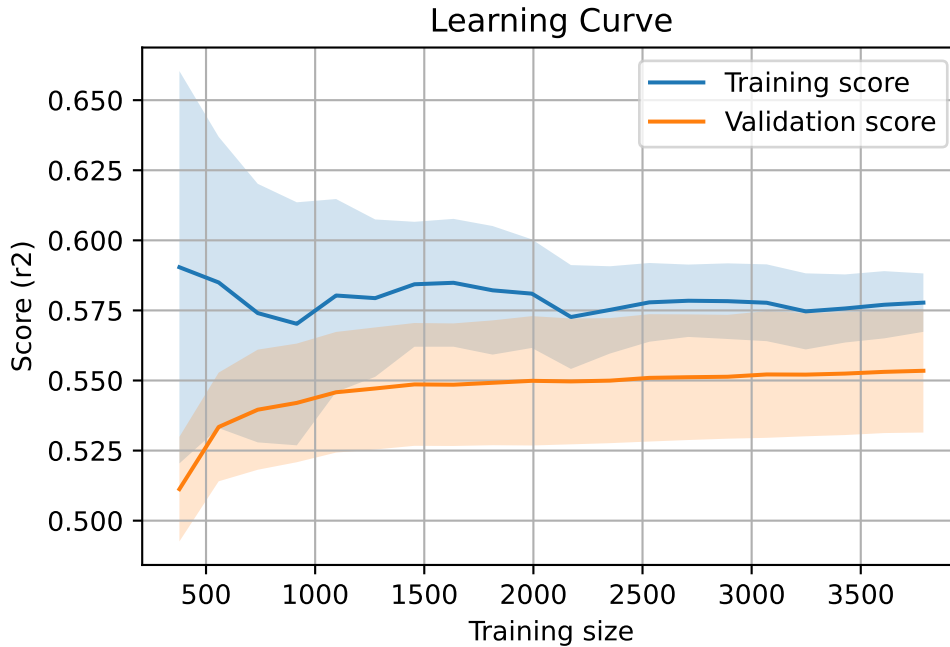
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



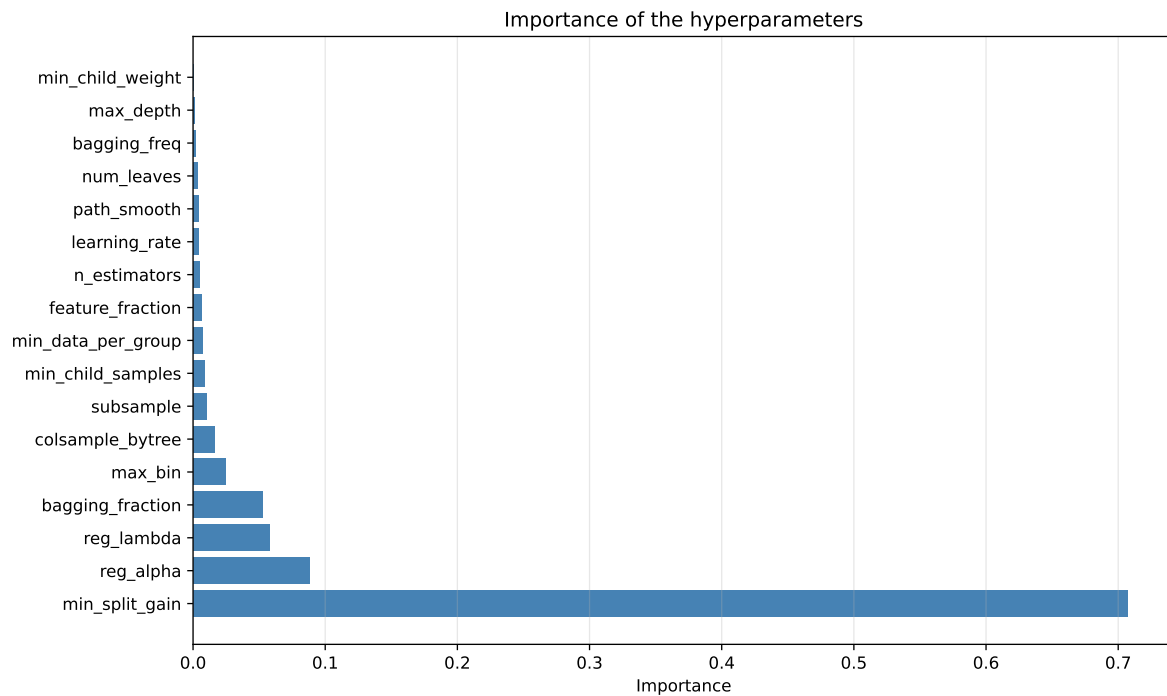
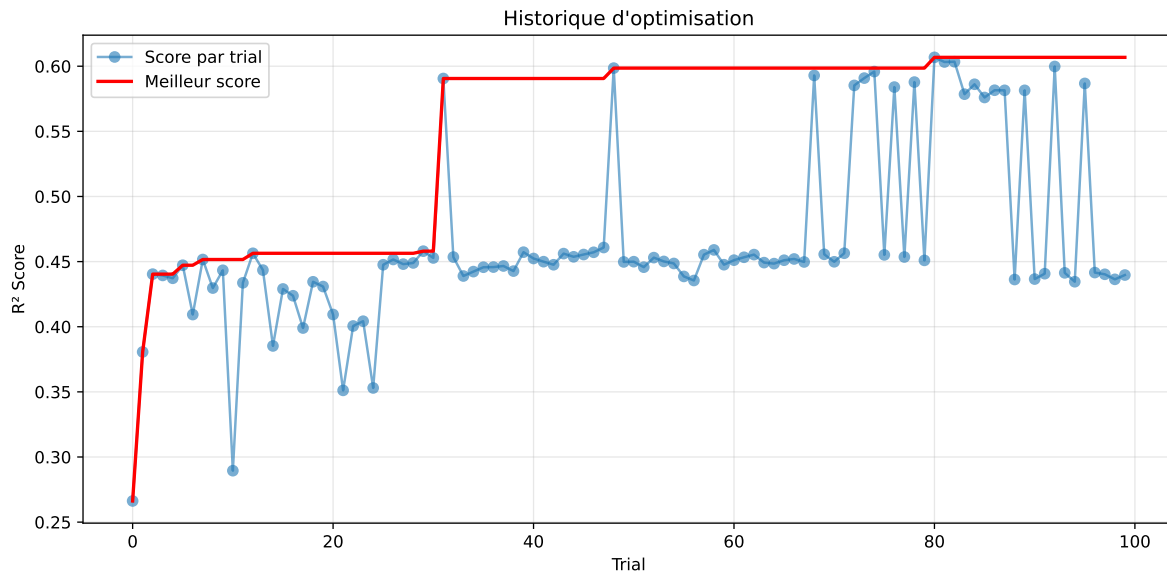
3.5 Model : LGBM

3.5.1 Gridsearch

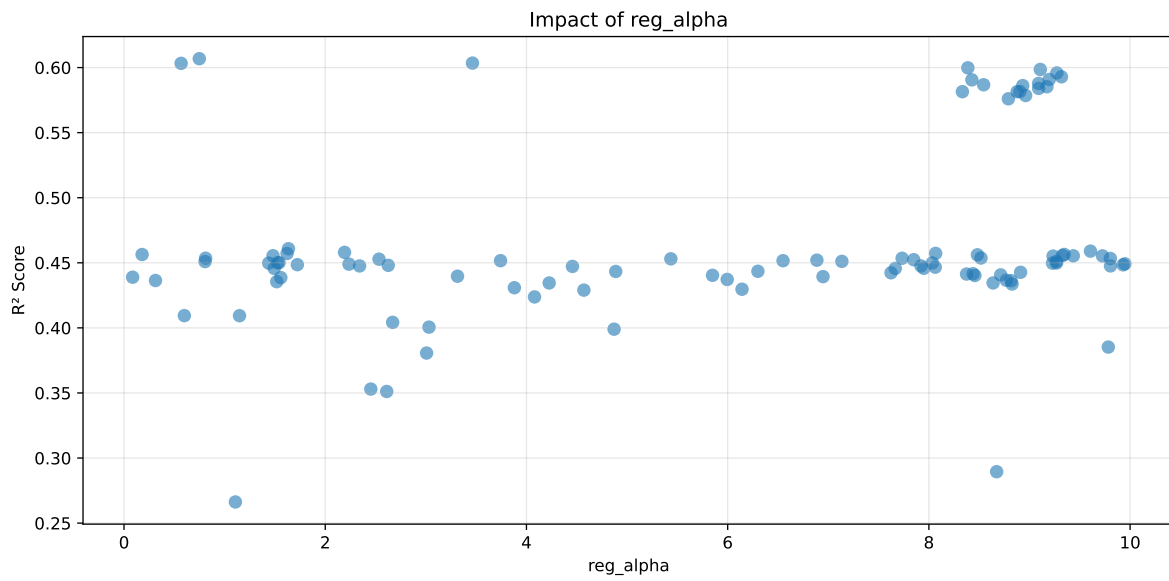
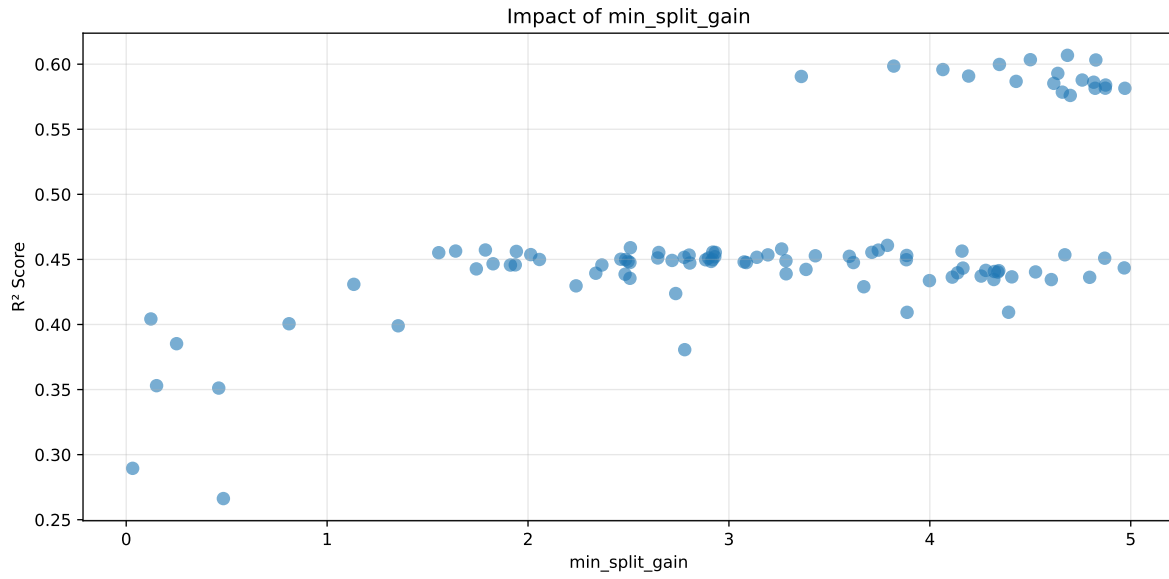
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

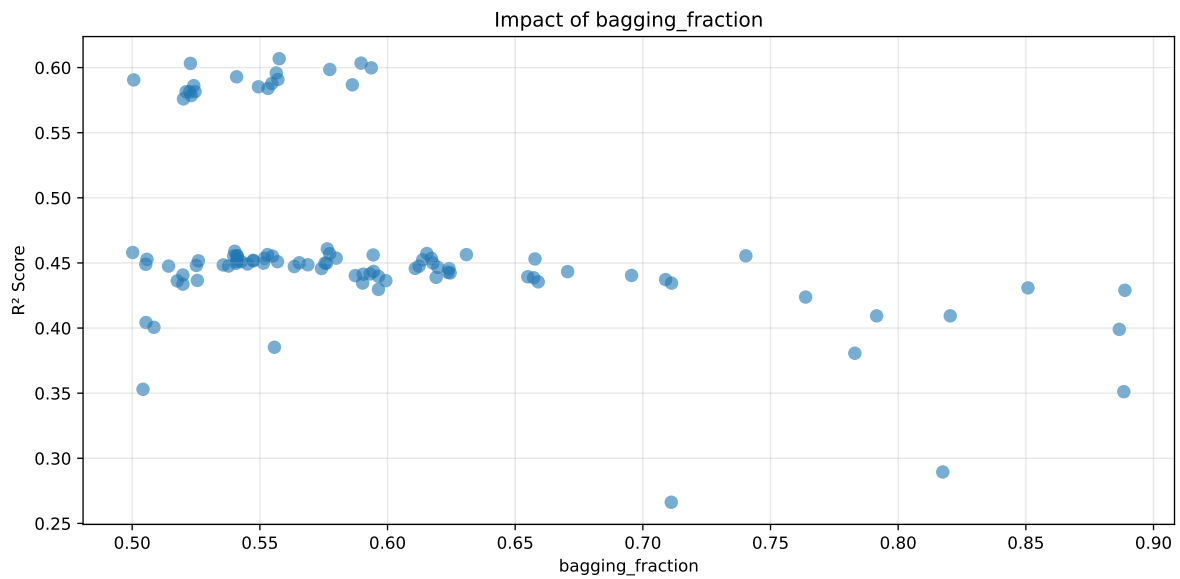
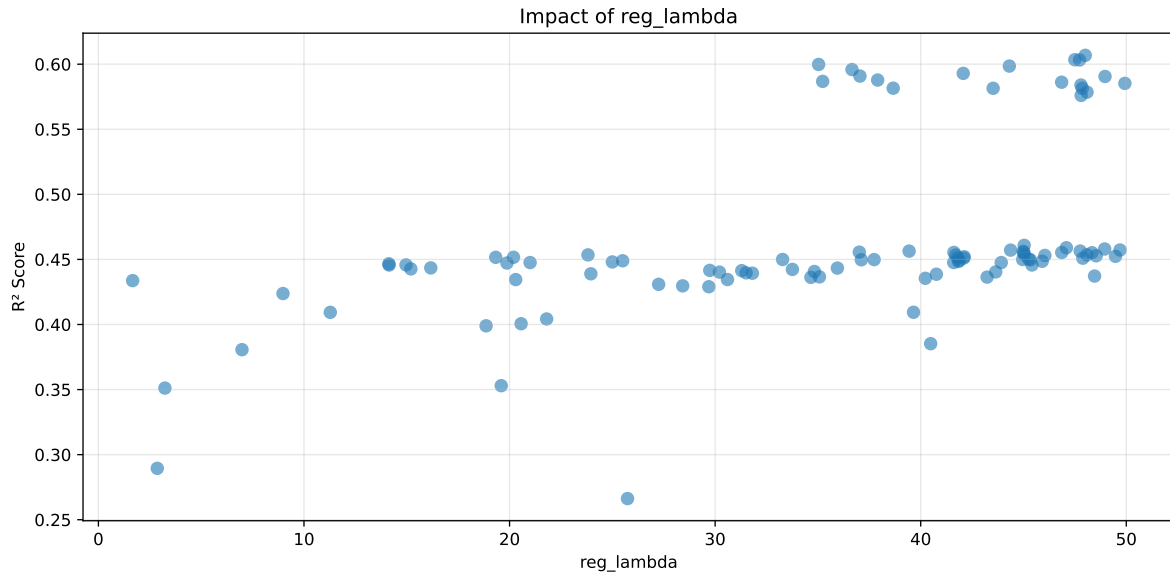
===== TOP Importance FEATURES =====

feature importance r_CO2 69 r_umidity 67 r_NH3_st_2h_std 50 r_temperature_st_2h_std
49 r_temperature_st_2h_diff_last 36 r_NH3_st_12h_mean 35 r_umidity_st_2h_slope 33
r_temperature 32 r_umidity_st_12h_mean 27 r_umidity_st_8h_max 25 r_NH3_st_8h_std
21 r_temperature_st_8h_slope 21 r_NH3 17 r_umidity_st_12h_slope 17 r_tempera-
ture_st_12h_min 17 r_umidity_st_12h_std 16 r_umidity_st_2h_diff_last 13 r_tempera-
ture_st_8h_diff_last 12 r_NH3_st_8h_slope 10 r_NH3_st_8h_min 9 r_NH3_st_2h_slope
9 r_THI 8 r_temperature_st_2h_mean 7 r_umidity_st_8h_std 7 r_NH3_st_2h_diff_last
7 r_NH3_st_2h_min 7 r_temperature_st_12h_diff_last 7 r_NH3_st_8h_diff_last
6 r_NH3_st_12h_slope 6 r_umidity_st_2h_std 6 r_umidity_st_8h_diff_last 5
r_temperature_st_12h_std 2 r_NH3_st_12h_std 2 r_NH3_st_12h_diff_last 1 r_umid-
ity_st_12h_diff_last 1



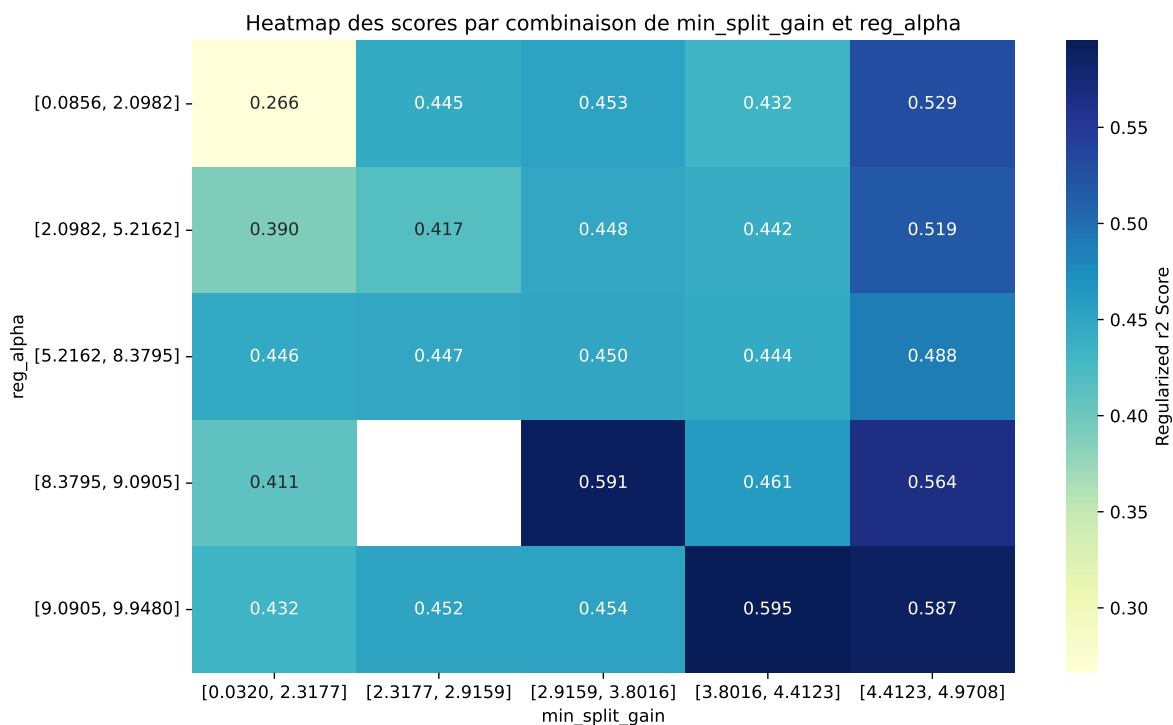
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'reg_lambda', 'bagging_fraction']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6354	0.6068	0.6492	0.0285	0.6068	0.6709	0.0641	1.1056	1.0472
2	0.6333	0.6034	0.647	0.0287	0.6034	0.673	0.0695	1.1152	1.0494
3	0.6312	0.6032	0.6453	0.0279	0.6032	0.668	0.0648	1.1074	1.0464
4	0.6293	0.5998	0.642	0.0296	0.5998	0.668	0.0682	1.1138	1.0491
5	0.6271	0.5985	0.6406	0.0263	0.5985	0.6582	0.0597	1.0997	1.0477
6	0.6255	0.5959	0.6392	0.0275	0.5959	0.6663	0.0705	1.1183	1.0498
7	0.6225	0.5929	0.637	0.0267	0.5929	0.662	0.0691	1.1165	1.0499
8	0.619	0.5909	0.6344	0.0269	0.5909	0.6579	0.067	1.1135	1.0476
9	0.6183	0.5906	0.6334	0.0288	0.5906	0.659	0.0684	1.1158	1.0469
10	0.616	0.5878	0.6298	0.032	0.5878	0.6612	0.0734	1.1249	1.048
---	---	---	---	---	---	---	---	---	---
100	0.8298	0.7359	0.7837	0.0316	0.2662	0.8043	0.0685	1.093	1.1276

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.03431047638429482, 'reg__num_leaves': 52, 'reg__subsample': 0.7387352892265153, 'reg__colsample_bytree': 0.6763991006279076, 'reg__min_child_samples': 21, 'reg__reg_alpha': 0.7480375534736416, 'reg__reg_lambda': 48.004703971029514, 'reg__min_split_gain': 4.6849656532225765, 'reg__path_smooth': 4.562471433901413, 'reg__min_child_weight':

0.04940740155424926, 'reg__feature_fraction': 0.5462073083563636, 'reg__bagging_fraction': 0.5575730962753629, 'reg__bagging_freq': 3, 'reg__max_bin': 75, 'reg__min_data_per_group': 158}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.01215224897033335, 'reg__num_leaves': 44, 'reg__subsample': 0.7412488356782558, 'reg__colsample_bytree': 0.6674710253989119, 'reg__min_child_samples': 21, 'reg__reg_alpha': 3.4633328252191173, 'reg__reg_lambda': 47.495415337104475, 'reg__min_split_gain': 4.500363477906915, 'reg__path_smooth': 1.8967801707578564, 'reg__min_child_weight': 0.04406250713338933, 'reg__feature_fraction': 0.6470374647948922, 'reg__bagging_fraction': 0.5896313765436844, 'reg__bagging_freq': 2, 'reg__max_bin': 74, 'reg__min_data_per_group': 154}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.034468743744312906, 'reg__num_leaves': 52, 'reg__subsample': 0.6457380454949349, 'reg__colsample_bytree': 0.6741428591493225, 'reg__min_child_samples': 21, 'reg__reg_alpha': 0.5673390290046056, 'reg__reg_lambda': 47.728813517807694, 'reg__min_split_gain': 4.826294635274093, 'reg__path_smooth': 2.061405515515416, 'reg__min_child_weight': 0.06480812890590631, 'reg__feature_fraction': 0.555145755701679, 'reg__bagging_fraction': 0.5228066831737362, 'reg__bagging_freq': 3, 'reg__max_bin': 76, 'reg__min_data_per_group': 157}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.03430537359708282, 'reg__num_leaves': 33, 'reg__subsample': 0.647602119852818, 'reg__colsample_bytree': 0.6995757828789079, 'reg__min_child_samples': 21, 'reg__reg_alpha': 8.386999379866639, 'reg__reg_lambda': 35.03370256241547, 'reg__min_split_gain': 4.34635003630913, 'reg__path_smooth': 1.2033224376525453, 'reg__min_child_weight': 0.35323590747374123, 'reg__feature_fraction': 0.6517799573724099, 'reg__bagging_fraction': 0.5936637580725097, 'reg__bagging_freq': 3, 'reg__max_bin': 152, 'reg__min_data_per_group': 156}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.012972586122853525, 'reg__num_leaves': 58, 'reg__subsample': 0.7181464936942653, 'reg__colsample_bytree': 0.800479010180533, 'reg__min_child_samples': 19, 'reg__reg_alpha': 9.107585644593552, 'reg__reg_lambda': 44.31179826053726, 'reg__min_split_gain': 3.8205062518427146, 'reg__path_smooth': 3.063887297179667, 'reg__min_child_weight': 0.15718458914266792, 'reg__feature_fraction': 0.5613931103541745, 'reg__bagging_fraction': 0.577382273790832, 'reg__bagging_freq': 2, 'reg__max_bin': 98, 'reg__min_data_per_group': 180}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.014905939317497856, 'reg__num_leaves': 36, 'reg__subsample': 0.6803643807185201, 'reg__colsample_bytree': 0.7134140401468971, 'reg__min_child_samples': 14, 'reg__reg_alpha': 9.272026582526555, 'reg__reg_lambda': 36.65789544440868, 'reg__min_split_gain': 4.064937414277537, 'reg__path_smooth': 2.408660686986936, 'reg__min_child_weight':

0.4272291850254335, 'reg__feature_fraction': 0.6218145943126421, 'reg__bagging_fraction': 0.5564331540894, 'reg__bagging_freq': 2, 'reg__max_bin': 116, 'reg__min_data_per_group': 189}

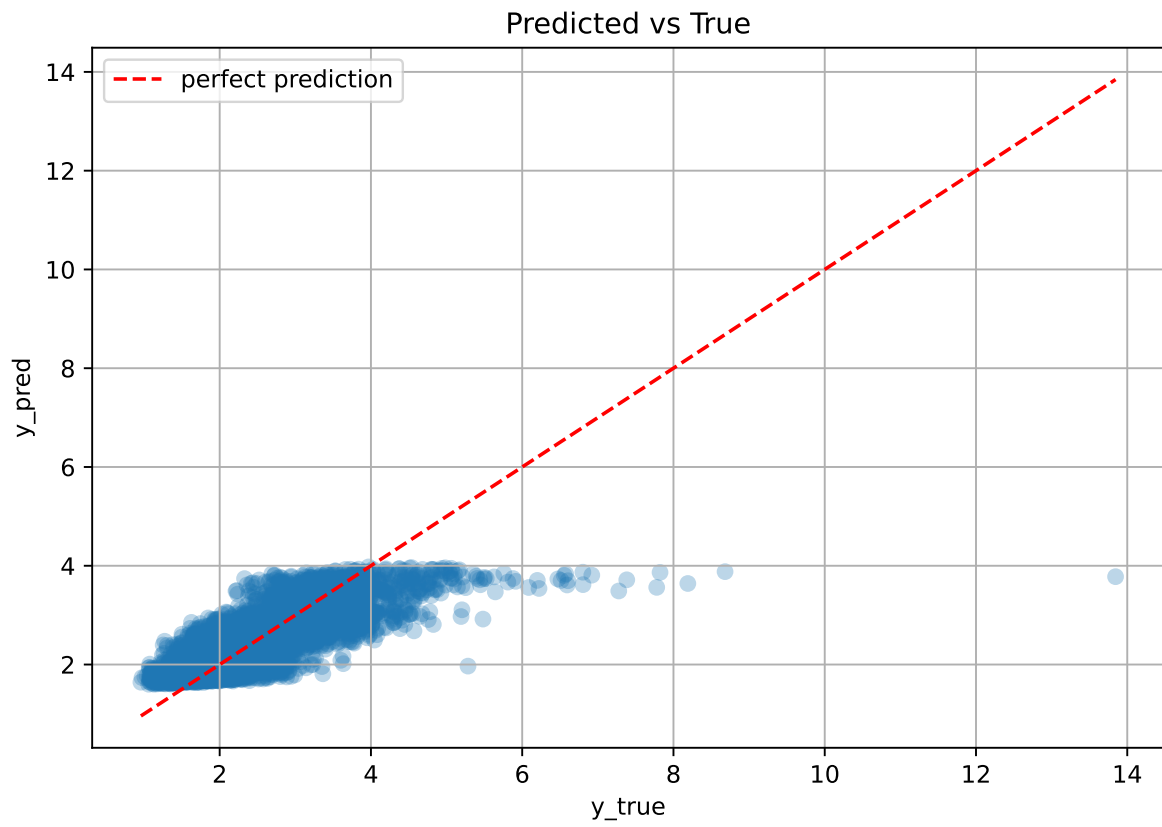
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.01618515843740525, 'reg__num_leaves': 54, 'reg__subsample': 0.7859191702113019, 'reg__colsample_bytree': 0.7264016843920391, 'reg__min_child_samples': 14, 'reg__reg_alpha': 9.317509076148827, 'reg__reg_lambda': 42.06873000015413, 'reg__min_split_gain': 4.636981307361397, 'reg__path_smooth': 2.851885466410543, 'reg__min_child_weight': 0.45063353460636557, 'reg__feature_fraction': 0.6133537996532815, 'reg__bagging_fraction': 0.5409014039377393, 'reg__bagging_freq': 1, 'reg__max_bin': 118, 'reg__min_data_per_group': 198}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.01801054033808842, 'reg__num_leaves': 35, 'reg__subsample': 0.6854497638984841, 'reg__colsample_bytree': 0.8280575493591477, 'reg__min_child_samples': 13, 'reg__reg_alpha': 9.19674174193042, 'reg__reg_lambda': 37.043953109693334, 'reg__min_split_gain': 4.19302560603487, 'reg__path_smooth': 4.02902462153701, 'reg__min_child_weight': 0.9893253175967165, 'reg__feature_fraction': 0.6184742156874408, 'reg__bagging_fraction': 0.5570723533697846, 'reg__bagging_freq': 2, 'reg__max_bin': 75, 'reg__min_data_per_group': 189}

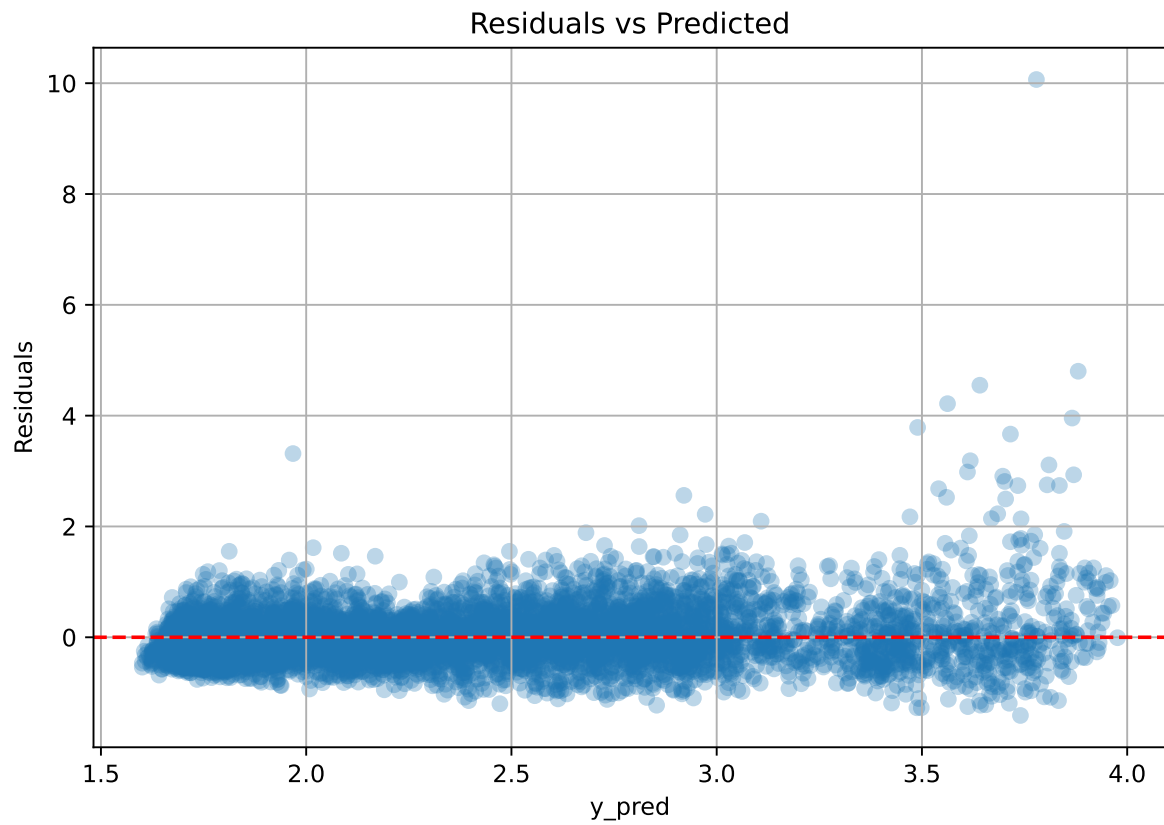
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.02458069769194249, 'reg__num_leaves': 48, 'reg__subsample': 0.7579013440936329, 'reg__colsample_bytree': 0.768267351475322, 'reg__min_child_samples': 22, 'reg__reg_alpha': 8.42733170381824, 'reg__reg_lambda': 48.96609918423635, 'reg__min_split_gain': 3.360397378246353, 'reg__path_smooth': 4.47211037450249, 'reg__min_child_weight': 0.013352846271992376, 'reg__feature_fraction': 0.7865110298726153, 'reg__bagging_fraction': 0.500597434399865, 'reg__bagging_freq': 3, 'reg__max_bin': 63, 'reg__min_data_per_group': 170}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.03435789076380272, 'reg__num_leaves': 53, 'reg__subsample': 0.6892166565369456, 'reg__colsample_bytree': 0.7601090989008817, 'reg__min_child_samples': 10, 'reg__reg_alpha': 9.090367478004826, 'reg__reg_lambda': 37.90682610562611, 'reg__min_split_gain': 4.758461825027579, 'reg__path_smooth': 1.9847880190076372, 'reg__min_child_weight': 0.37571018439129117, 'reg__feature_fraction': 0.6321549992499415, 'reg__bagging_fraction': 0.5546808261127777, 'reg__bagging_freq': 2, 'reg__max_bin': 152, 'reg__min_data_per_group': 189}

3.5.2 Scatter



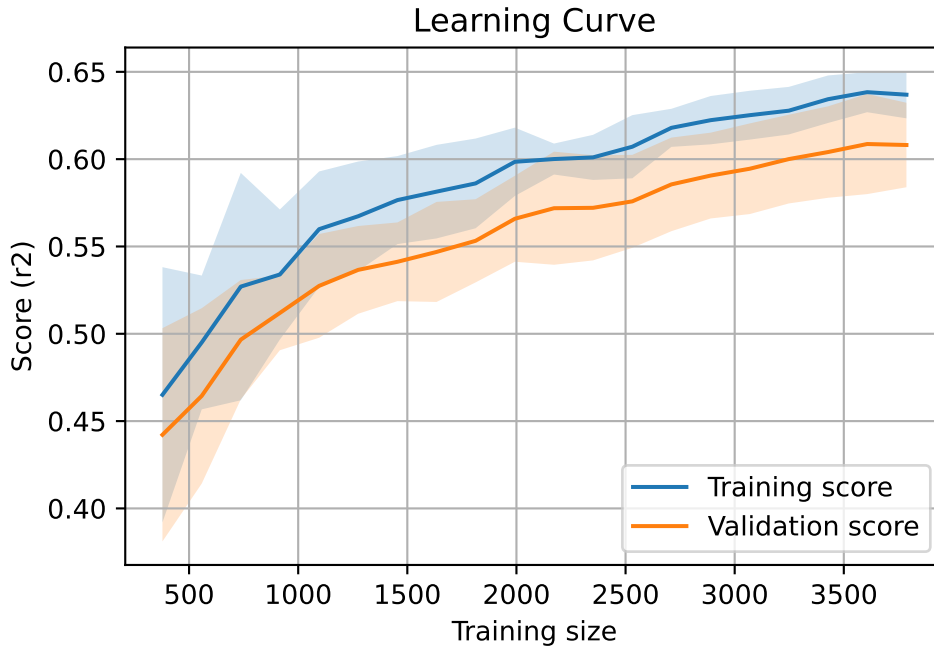
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



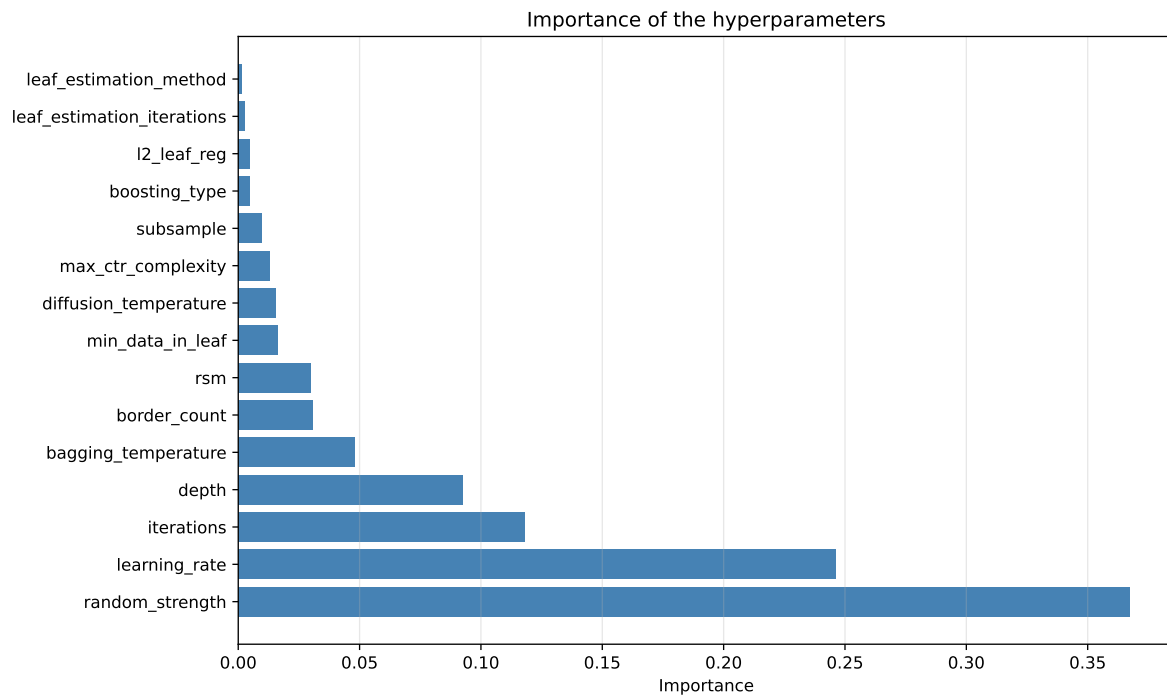
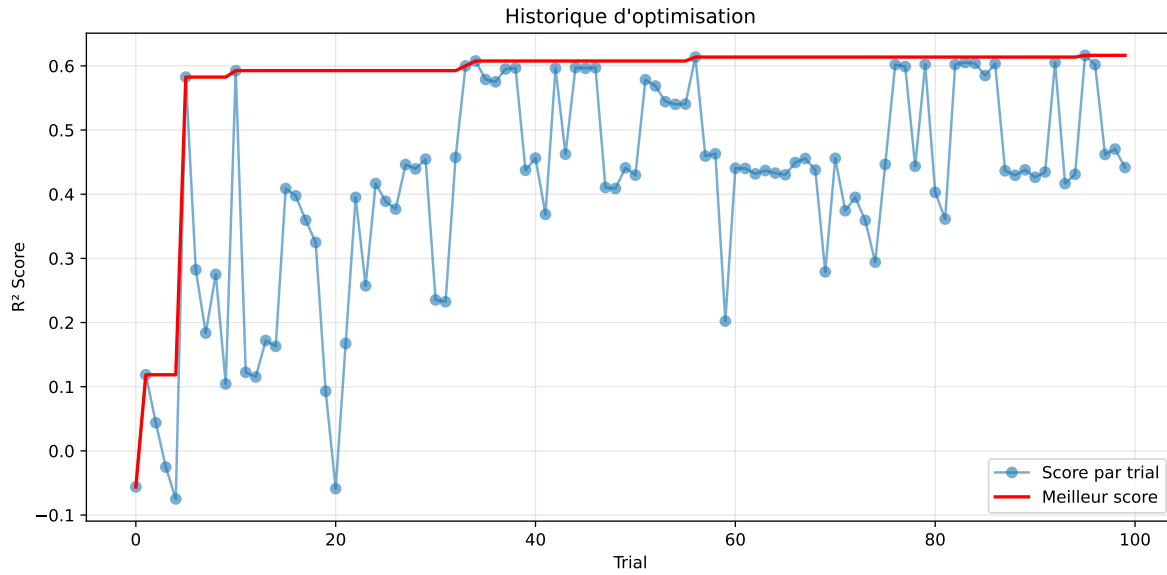
3.6 Model : CatBoost

3.6.1 Gridsearch

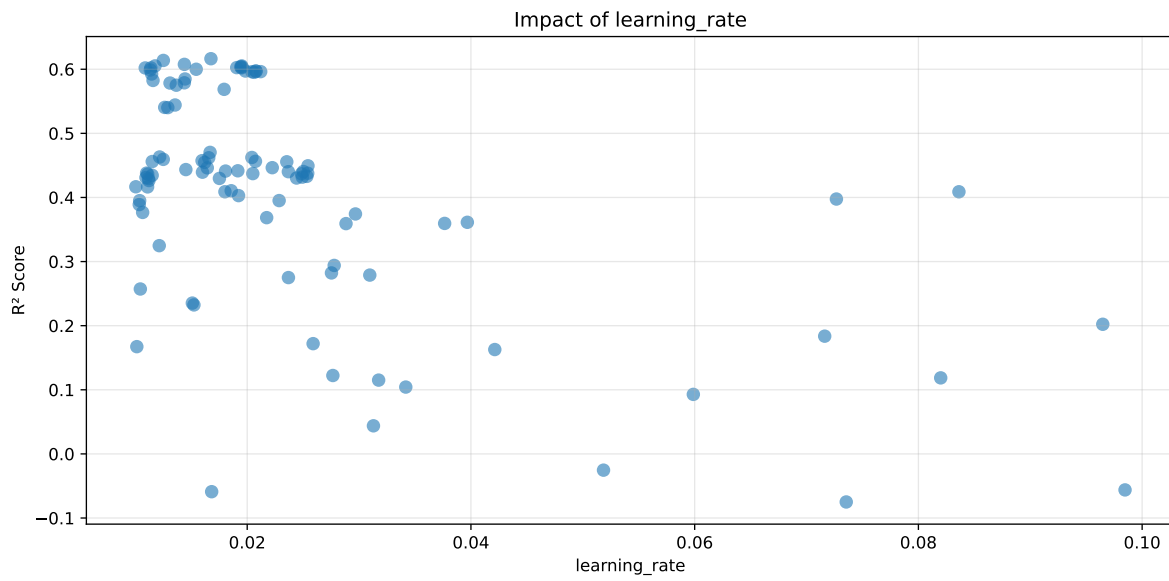
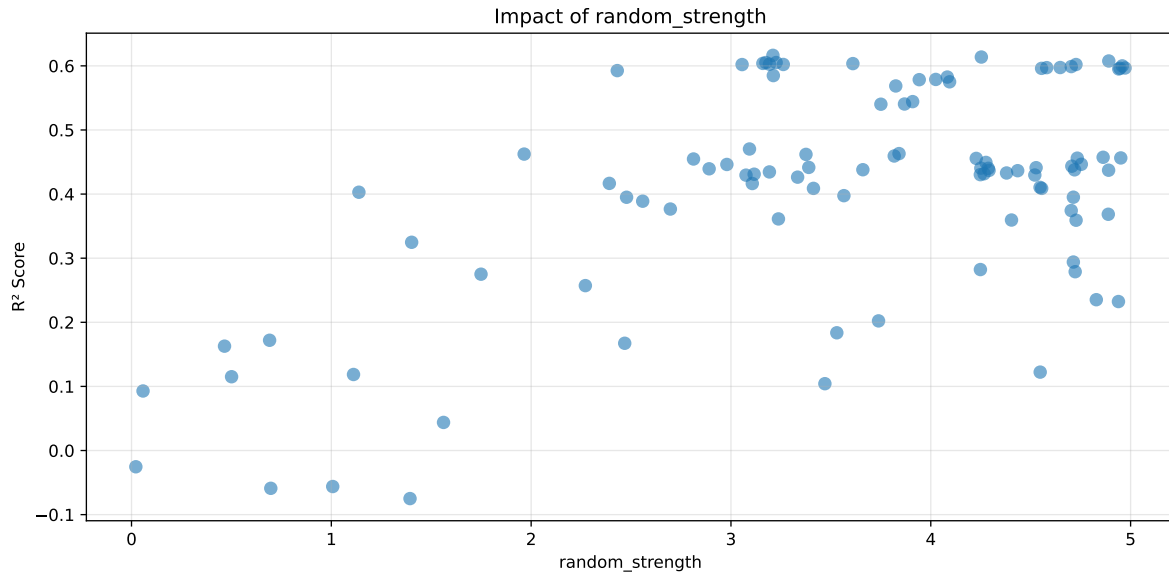
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

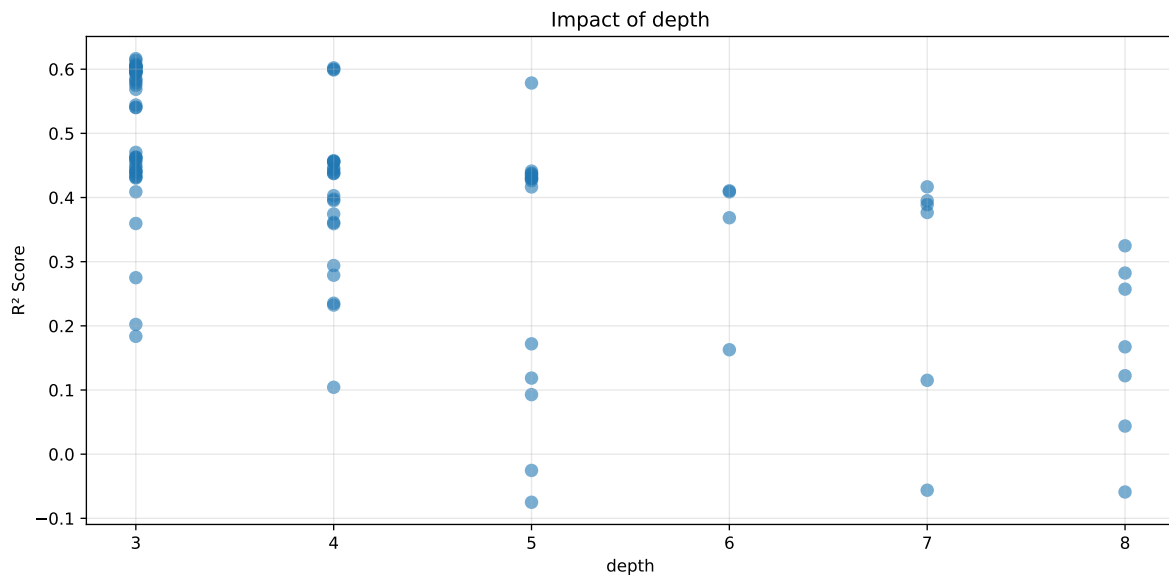
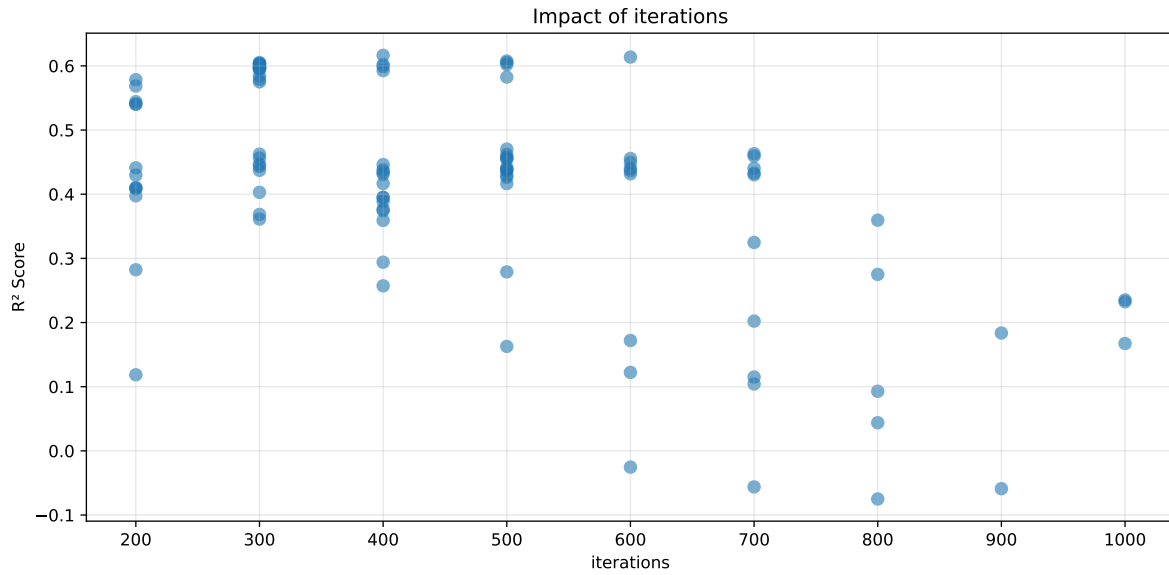
===== TOP Importance FEATURES =====

feature importance	r_umidity	r_temperature_st_2h_std	r_umidity_st_2h_slope	r_temperature_st_8h_slope	r_temperature_st_2h_diff_last
26.237530	12.389903	10.374831	6.568897	4.751960	3.718196
3.660588	2.816464	2.721312	2.277483	2.263564	1.843483
1.839385	1.839385	1.654378	1.512286	1.488268	1.425402
1.425402	1.317882	1.306600	1.253865	1.229159	1.142904
0.897068	0.744372	0.713520	0.664766	0.590237	0.442053
0.442053	0.402626	0.401821	0.320037	0.298413	0.258002
0.258002	0.247491	0.225255			



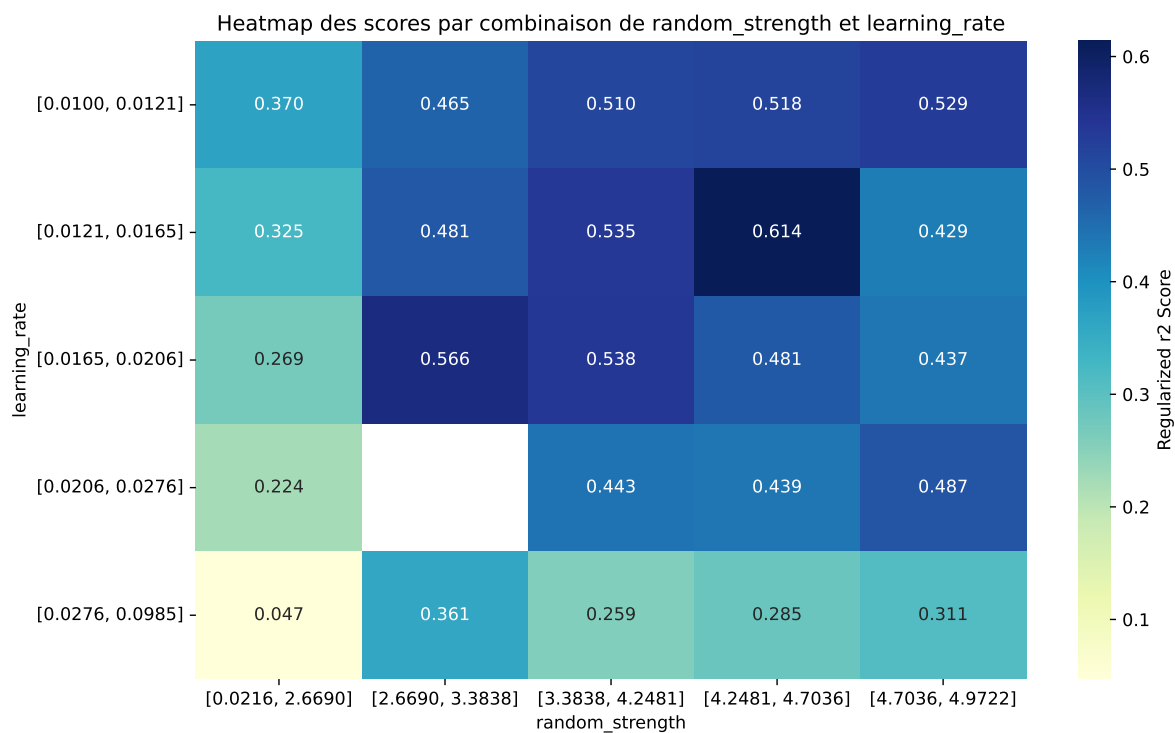
Paramètres avec >5% d'importance : ['random_strength', 'learning_rate', 'iterations', 'depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6471	0.6164	0.6629	0.0252	0.6164	0.6687	0.0523	1.0848	1.0497
2	0.642	0.6137	0.6605	0.0256	0.6137	0.6662	0.0525	1.0855	1.0461
3	0.6345	0.6076	0.6522	0.0245	0.6076	0.6562	0.0486	1.08	1.0442
4	0.6321	0.6052	0.6502	0.0253	0.6052	0.6545	0.0493	1.0815	1.0445
5	0.6317	0.6052	0.6491	0.0235	0.6052	0.6557	0.0505	1.0834	1.0438
6	0.6315	0.6038	0.6482	0.0251	0.6038	0.6536	0.0497	1.0823	1.0458
7	0.6301	0.6035	0.6479	0.0261	0.6035	0.6536	0.0501	1.083	1.0441
8	0.6295	0.6025	0.6469	0.0242	0.6025	0.6509	0.0483	1.0802	1.0448
9	0.6303	0.6022	0.6473	0.0249	0.6022	0.6538	0.0516	1.0858	1.0467
10	0.6309	0.6021	0.6465	0.0256	0.6021	0.6517	0.0496	1.0824	1.0479
---	---	---	---	---	---	---	---	---	---
100	0.9601	0.7876	0.8282	0.0285	-0.075	0.8423	0.0548	1.0696	1.219

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.016758391550041933, 'reg__l2_leaf_reg': 13.837279468885976, 'reg__bagging_temperature': 0.5476604618300113, 'reg__random_strength': 3.2100264718213545, 'reg__subsample': 0.8067646973713577, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8867835241377021, 'reg__border_count': 68, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8772.503498662947}

Rank 2: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.012505165770441607, 'reg__l2_leaf_reg': 15.108936087952035, 'reg__bagging_temperature': 2.914082794746181, 'reg__random_strength': 4.253062968870105, 'reg__subsample': 0.6002742125664471, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.785886751798466, 'reg__border_count': 69, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4630.593159519414}

Rank 3: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01438476148522337, 'reg__l2_leaf_reg': 7.5851063338983025, 'reg__bagging_temperature': 0.48018904835929493, 'reg__random_strength': 4.890852939272258, 'reg__subsample': 0.6614174435981909, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8782065104226329, 'reg__border_count': 63, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3009.9350737080017}

Rank 4: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011759617257319664, 'reg__l2_leaf_reg': 34.3709769121908, 'reg__bagging_temperature': 0.4593636081561696, 'reg__random_strength': 3.2260665126115136, 'reg__subsample': 0.7524221259698508, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8863324532491751, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4147.714023550769}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019509873390876974, 'reg__l2_leaf_reg': 17.16752517988449, 'reg__bagging_temperature': 0.8042651935051965, 'reg__random_strength': 3.1746681778382495, 'reg__subsample': 0.7917139996234822, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8619278774241153, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5114.746384592046}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019425869133030862, 'reg__l2_leaf_reg': 13.656553202978566, 'reg__bagging_temperature': 0.8302139514608038, 'reg__random_strength': 3.159562930783334, 'reg__subsample': 0.7513429106210783, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8590787462407476, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5053.290725442015}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019529359318706654, 'reg__l2_leaf_reg': 9.181549974391773, 'reg__bagging_temperature': 0.45307968423200884, 'reg__random_strength': 3.610092712331718, 'reg__subsample': 0.7802459356269865,

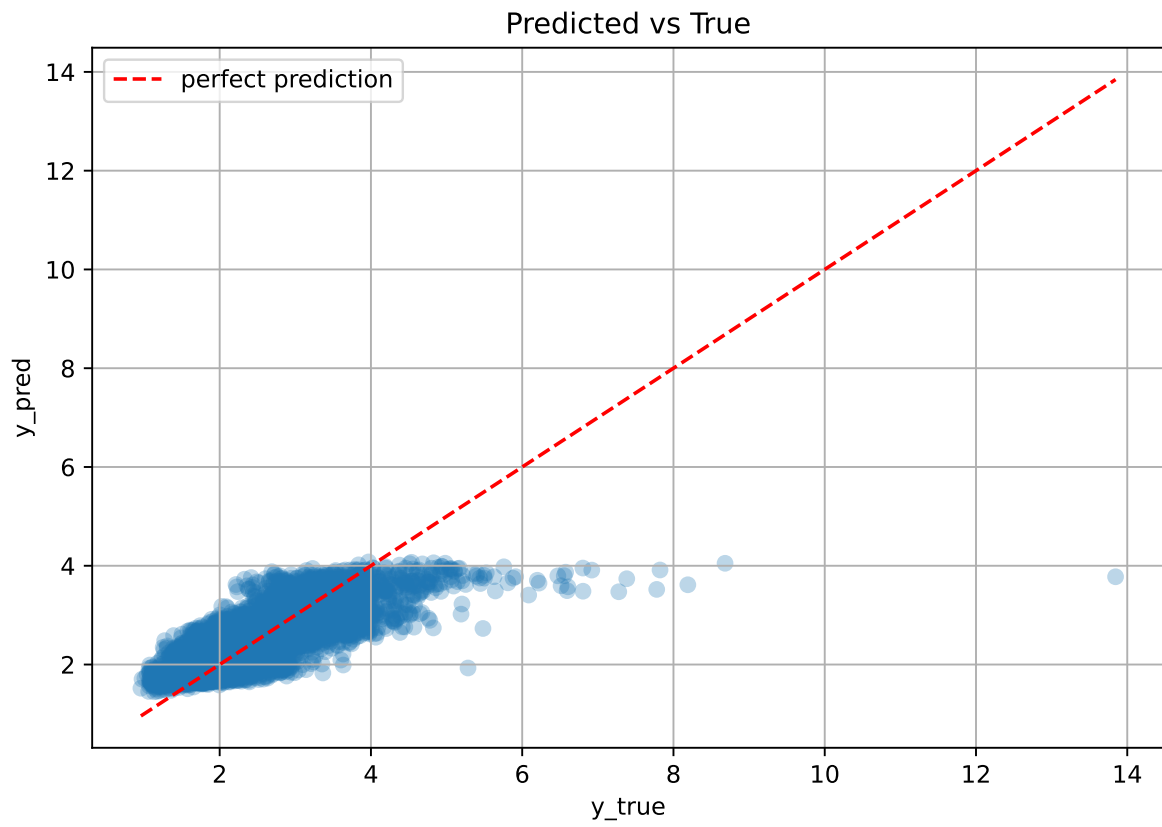
'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.861412787245985, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8781.361166502045}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019062605486600732, 'reg__l2_leaf_reg': 9.87412588836806, 'reg__bagging_temperature': 0.570289753735275, 'reg__random_strength': 3.192688511312764, 'reg__subsample': 0.7316295985716328, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8352207911665317, 'reg__border_count': 66, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5148.123680516437}

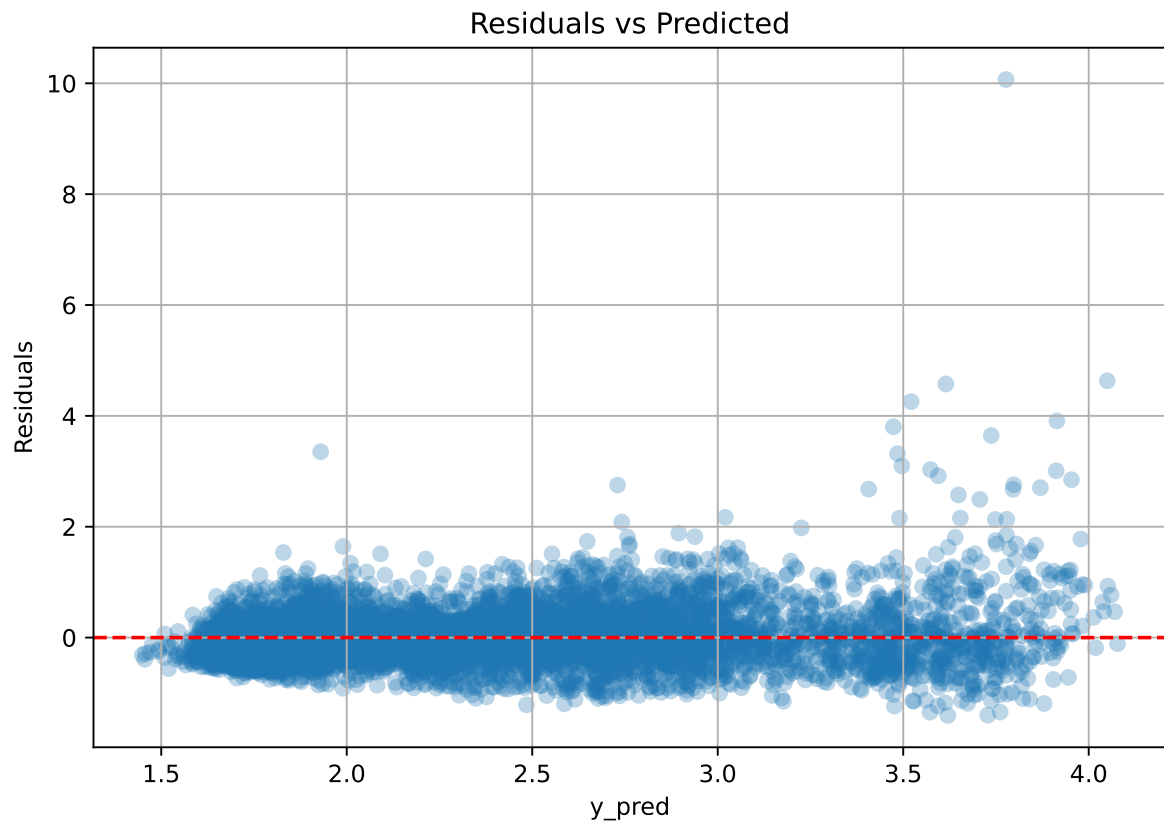
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01948486234667863, 'reg__l2_leaf_reg': 24.74219172973435, 'reg__bagging_temperature': 0.8219866775628755, 'reg__random_strength': 3.261894876789049, 'reg__subsample': 0.7457618973378637, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8366302319147915, 'reg__border_count': 66, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5138.613317098647}

Rank 10: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.011385060522105151, 'reg__l2_leaf_reg': 10.190668051036141, 'reg__bagging_temperature': 1.1440160620035056, 'reg__random_strength': 4.726662005464462, 'reg__subsample': 0.7317079242501019, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.891882244350692, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8252.72033576075}

3.6.2 Scatter



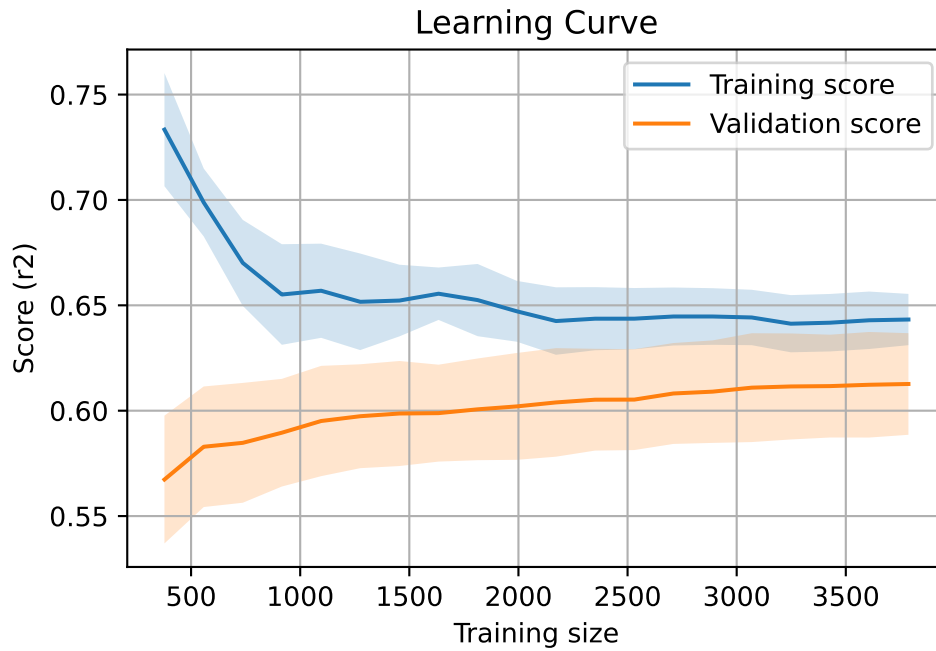
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.578	0.554	0.022	0.597	0.599	0.045	1.082	1.044	nan
LGBM	0.635	0.607	0.028	0.649	0.671	0.064	1.106	1.047	0.6068
CatBoost	0.647	0.616	0.025	0.663	0.669	0.052	1.085	1.05	0.6164

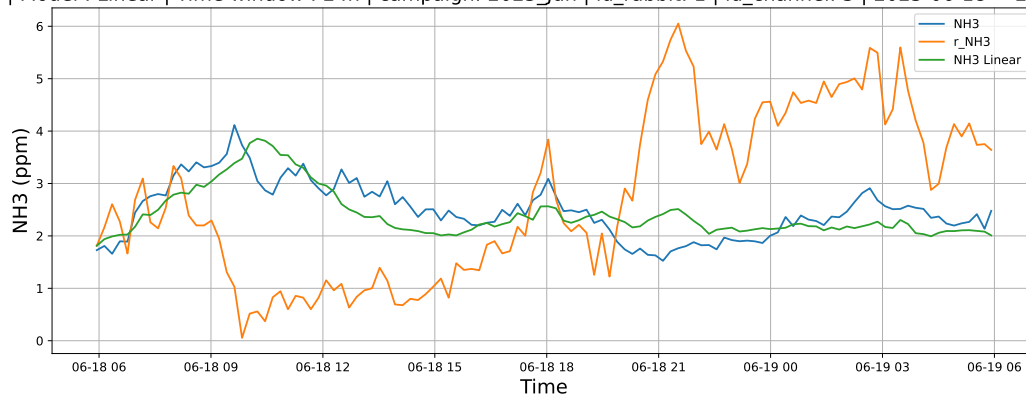
4 Plot

4.1 Standalone plots

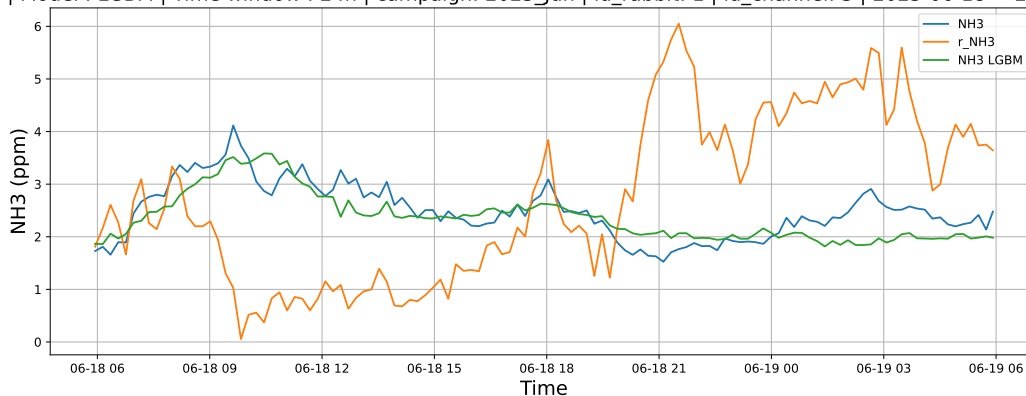
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

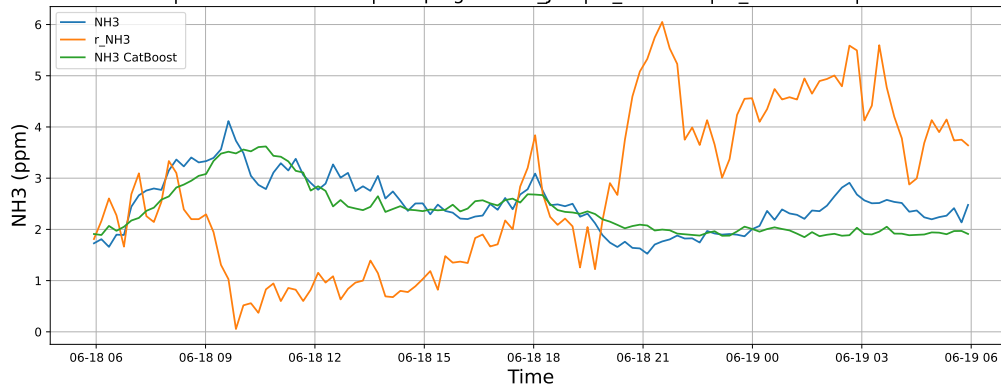
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

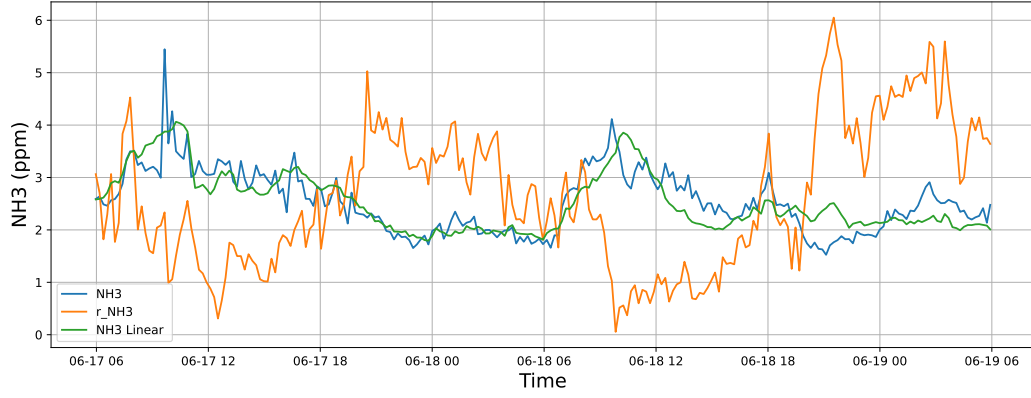


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

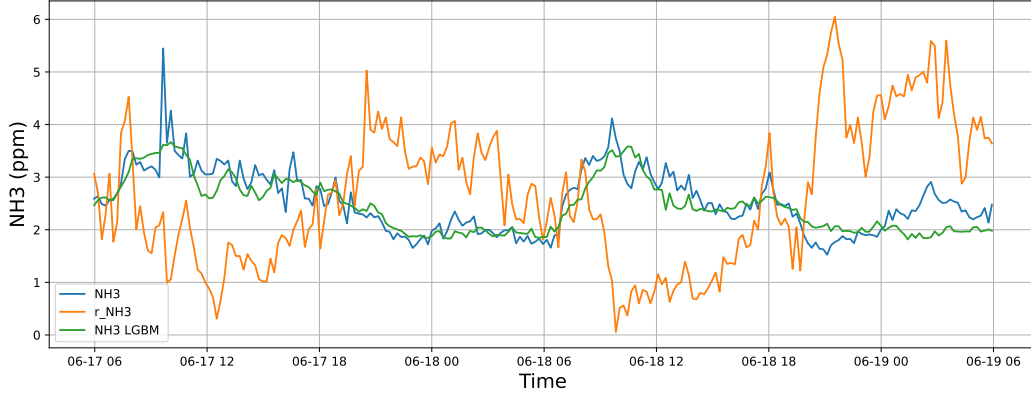


4.1.1.2 Time window : 48

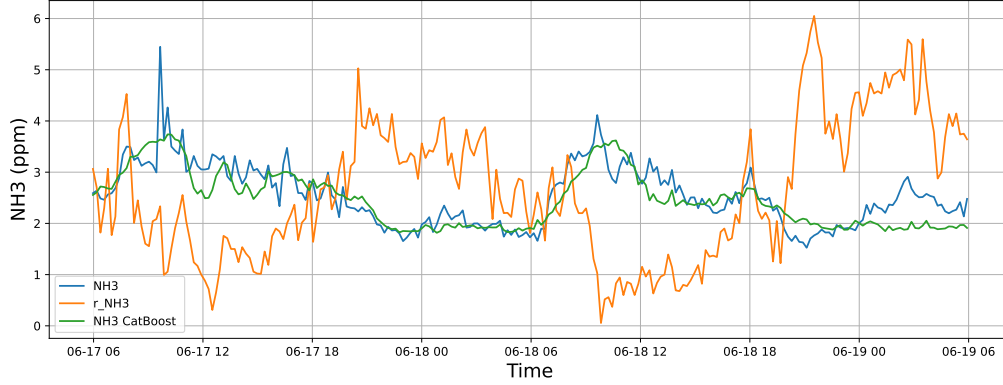
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

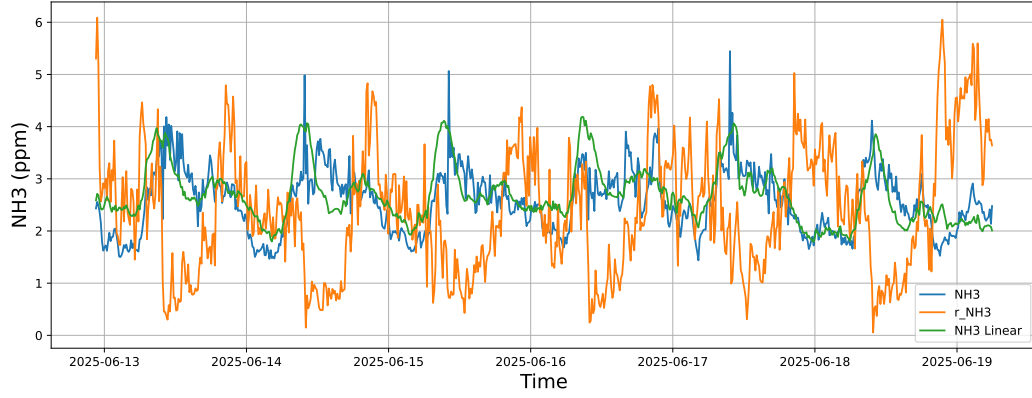


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

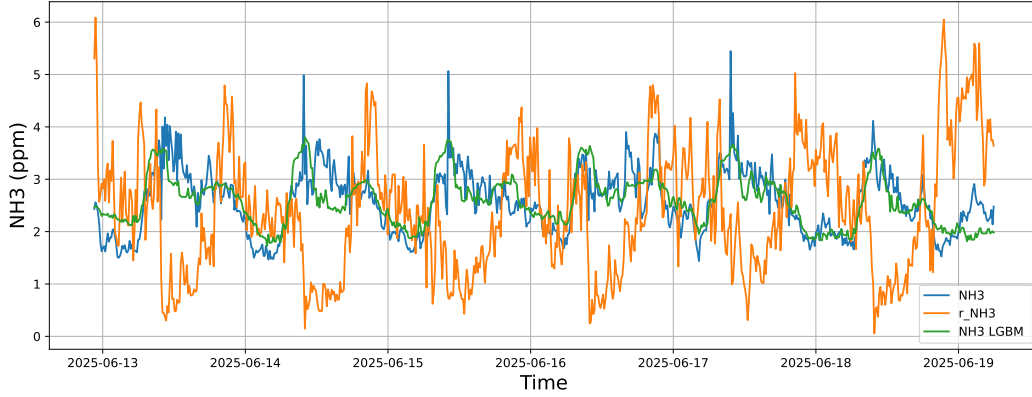


4.1.1.3 Time window : ALL

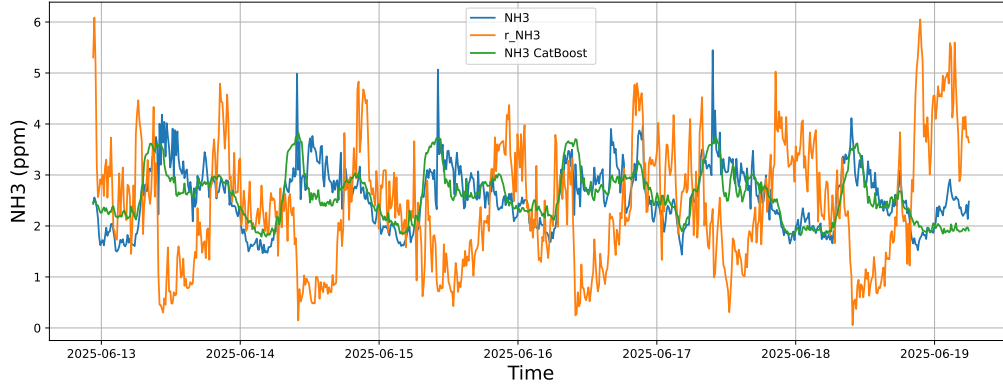
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



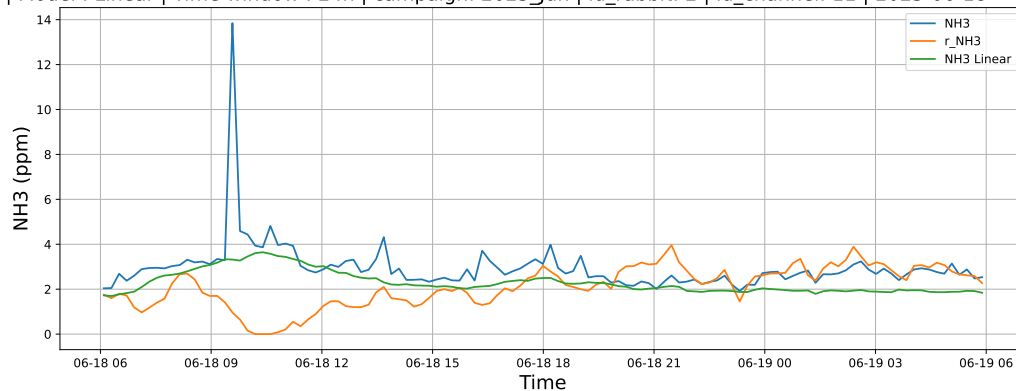
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



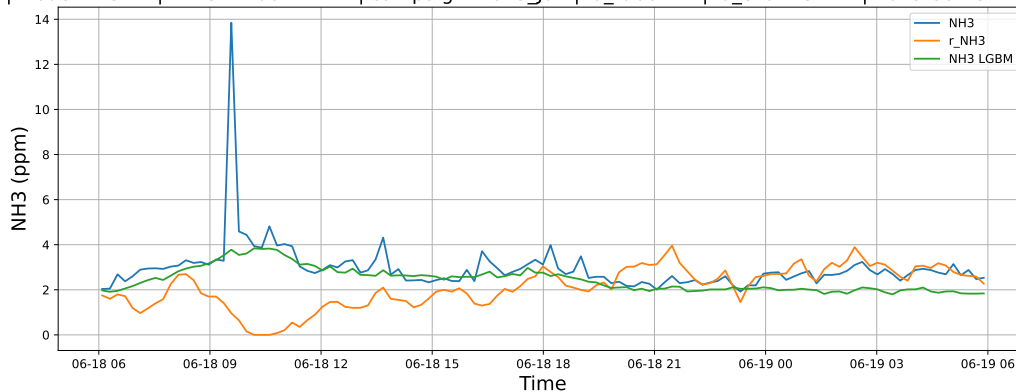
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

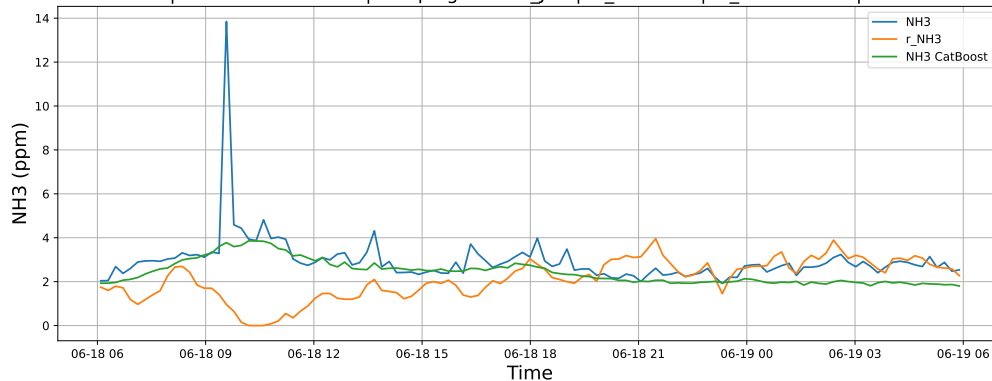
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

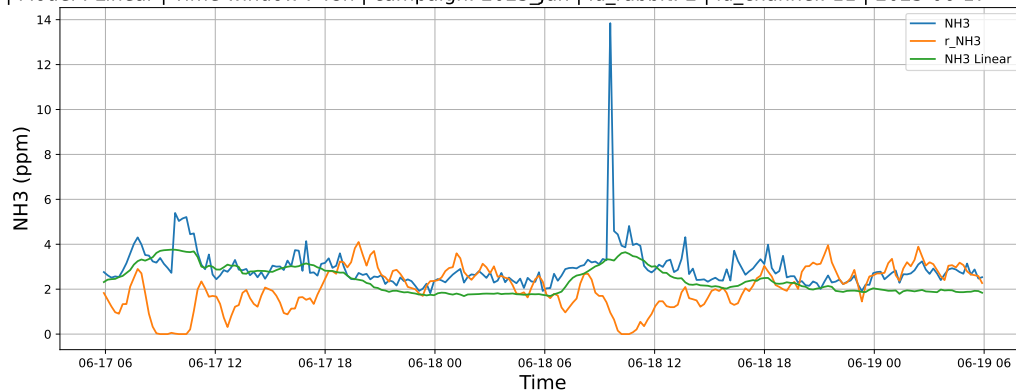


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

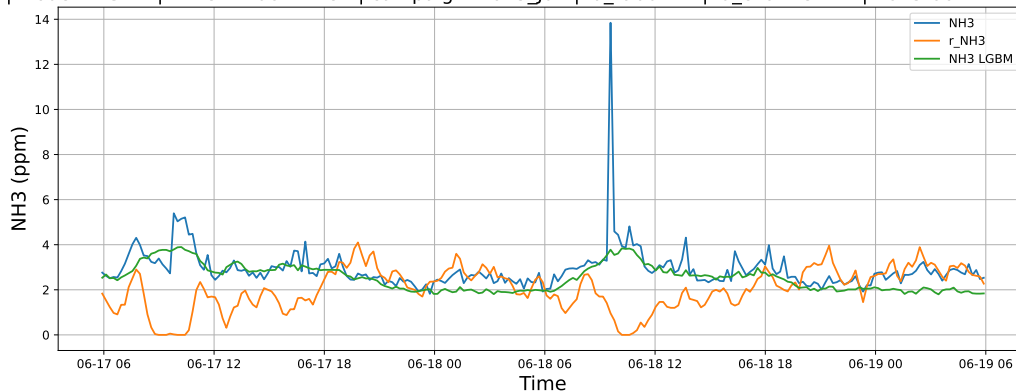


4.1.2.2 Time window : 48

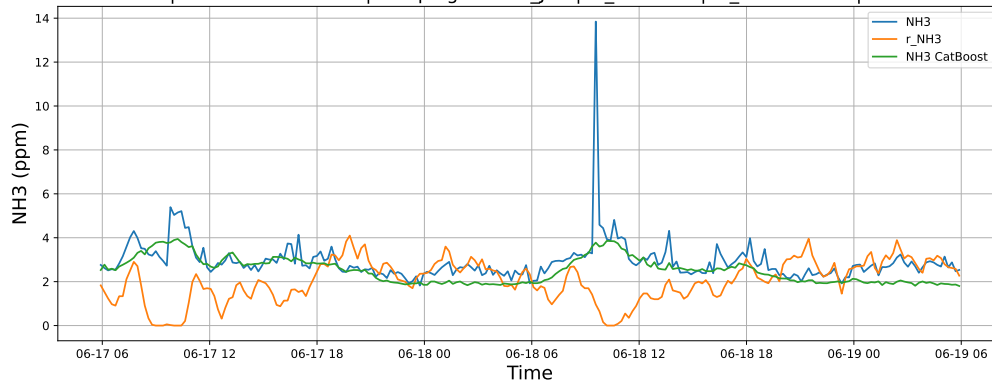
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

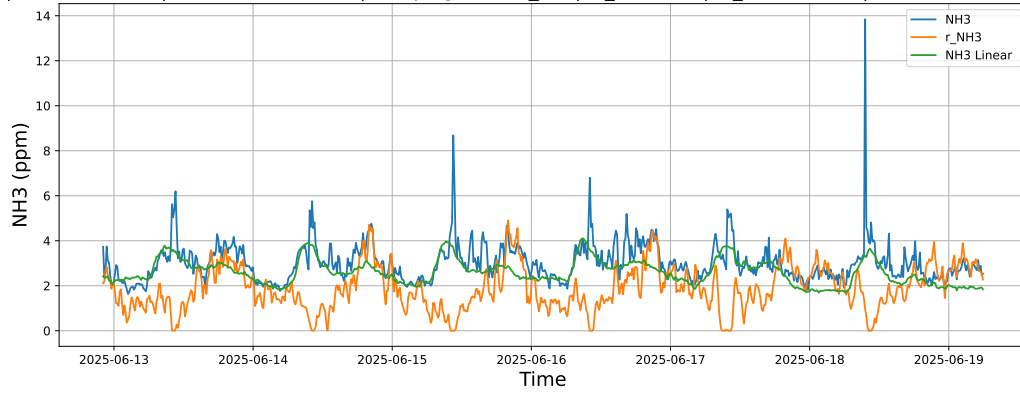


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

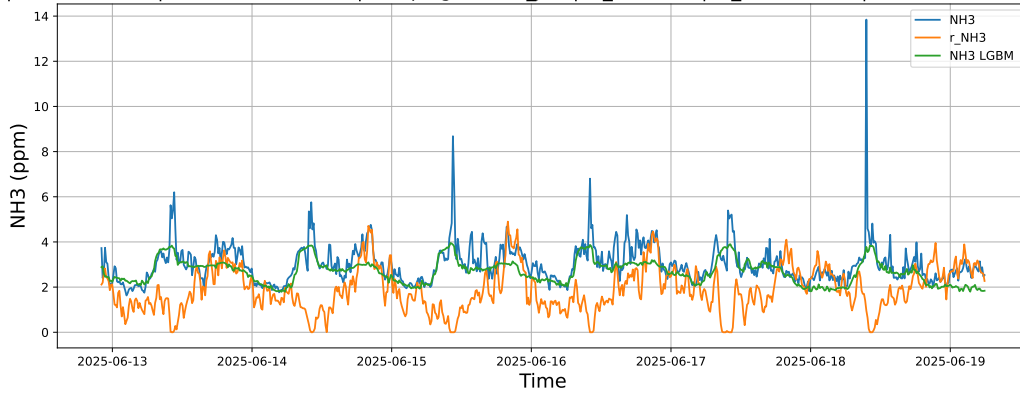


4.1.2.3 Time window : ALL

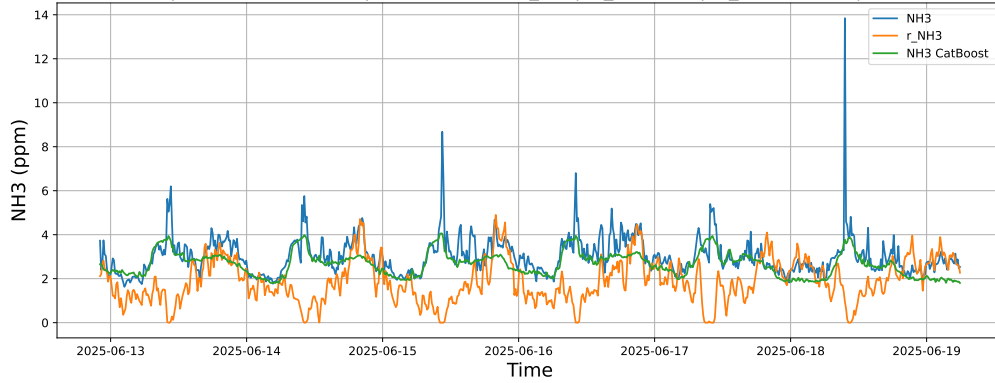
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



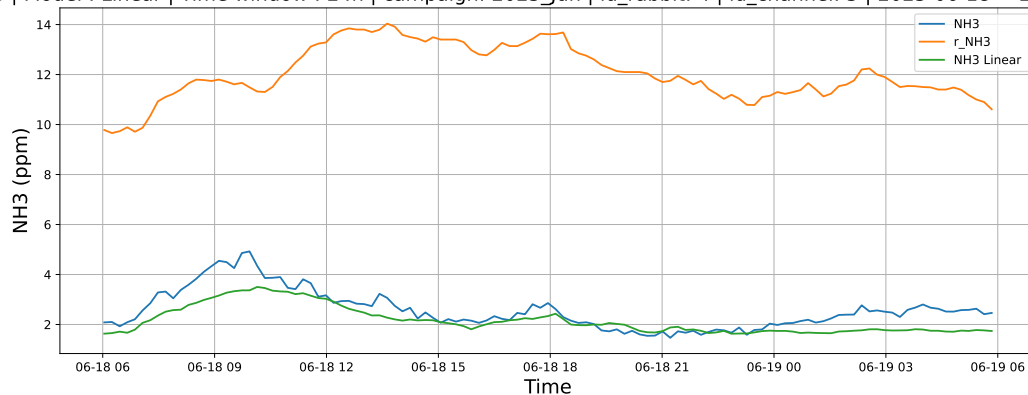
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



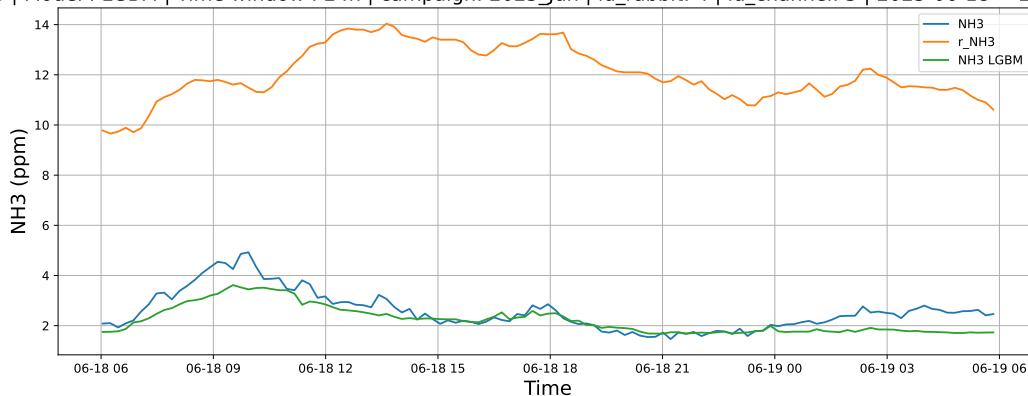
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

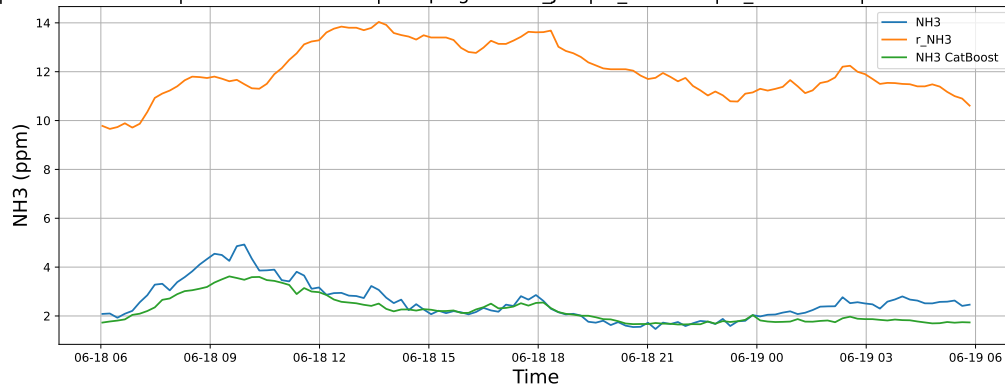
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

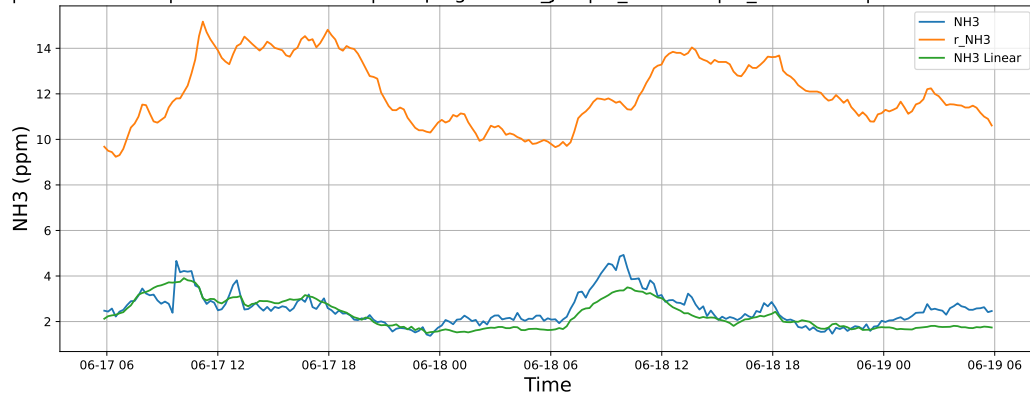


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

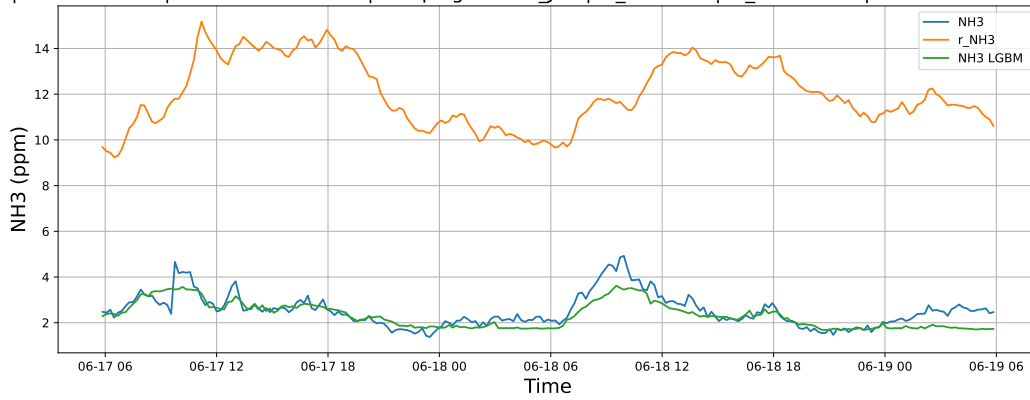


4.1.3.2 Time window : 48

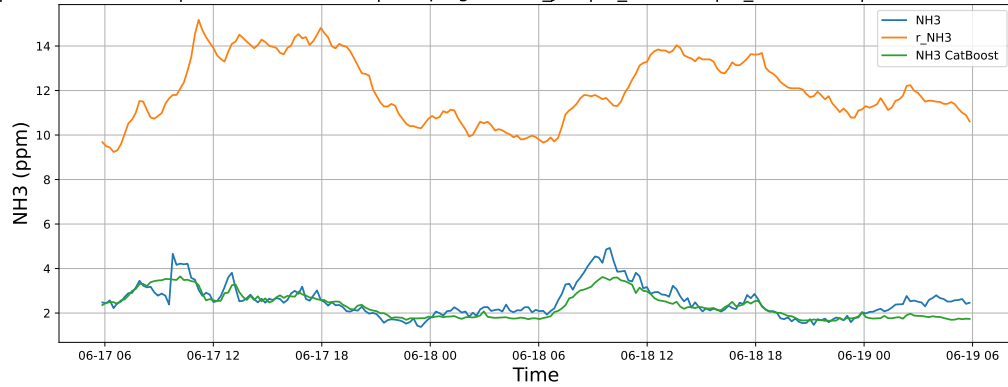
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

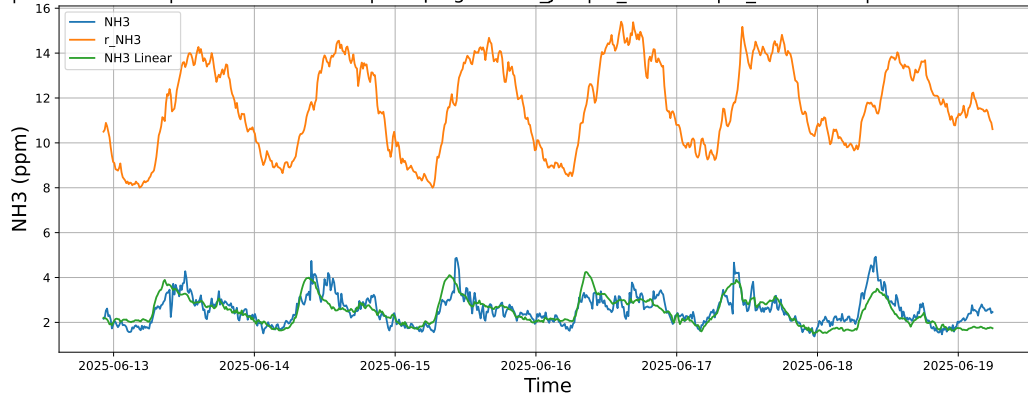


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

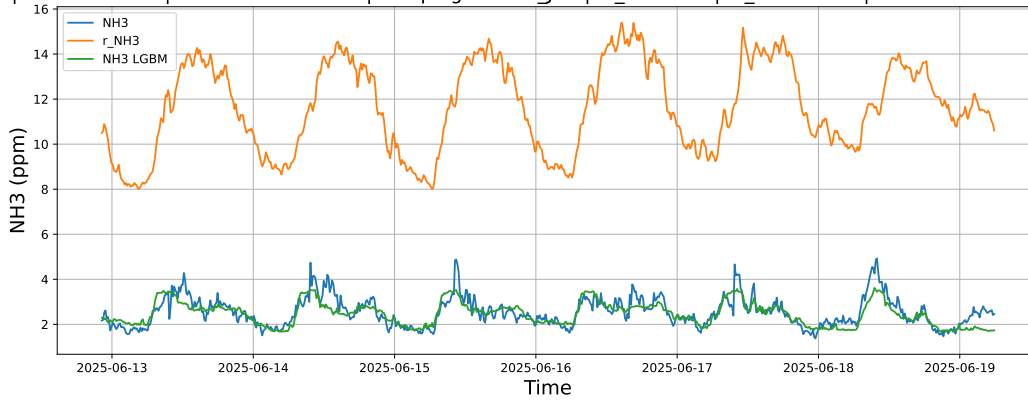


4.1.3.3 Time window : ALL

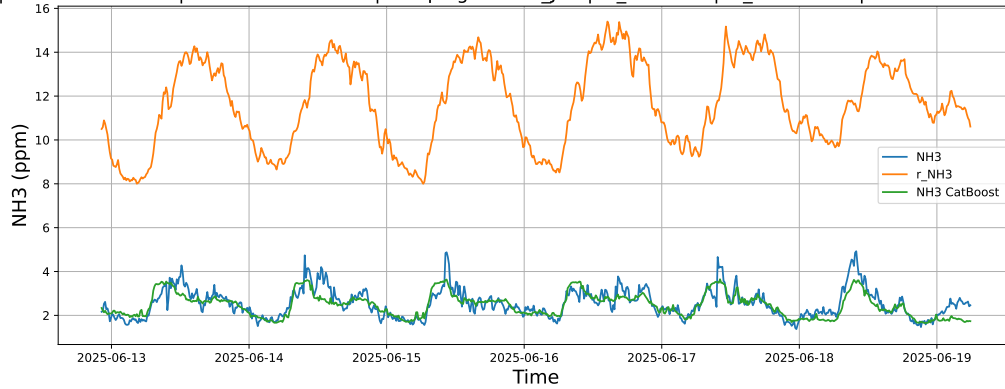
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



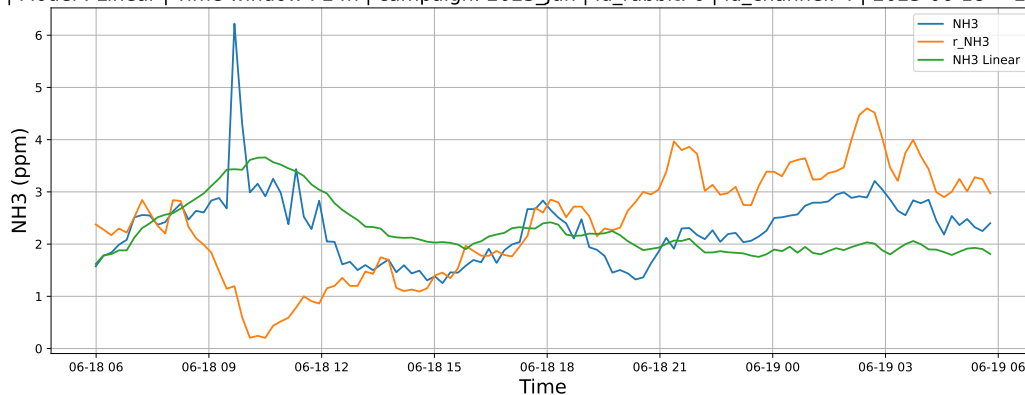
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



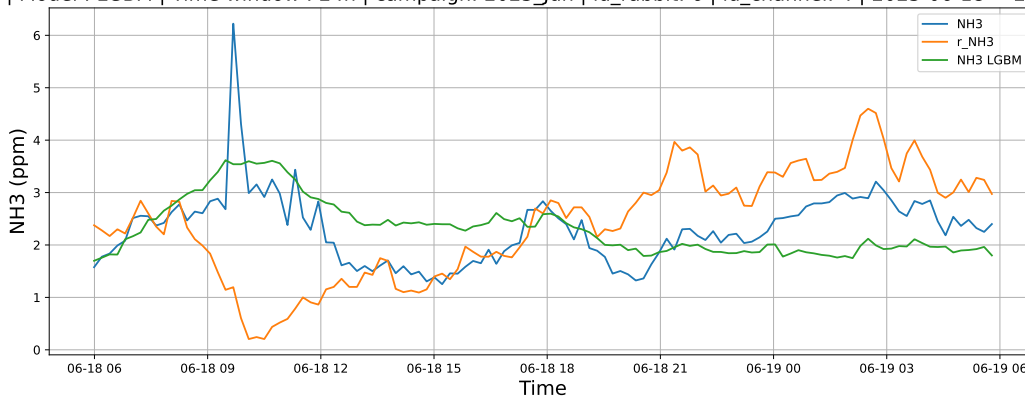
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

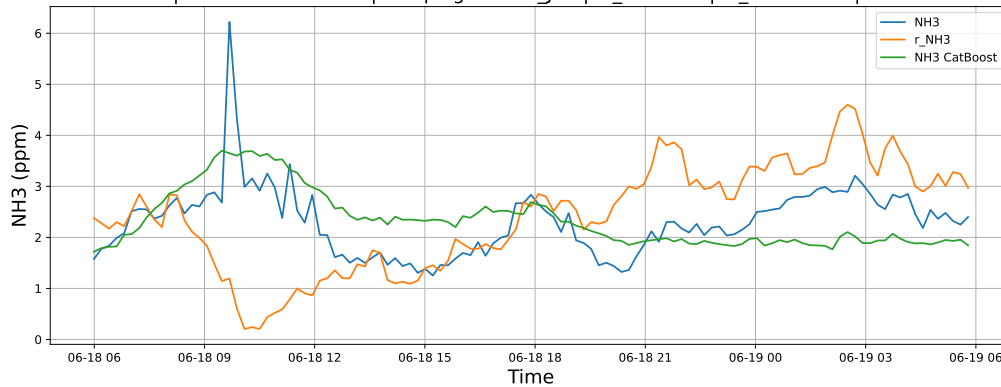
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

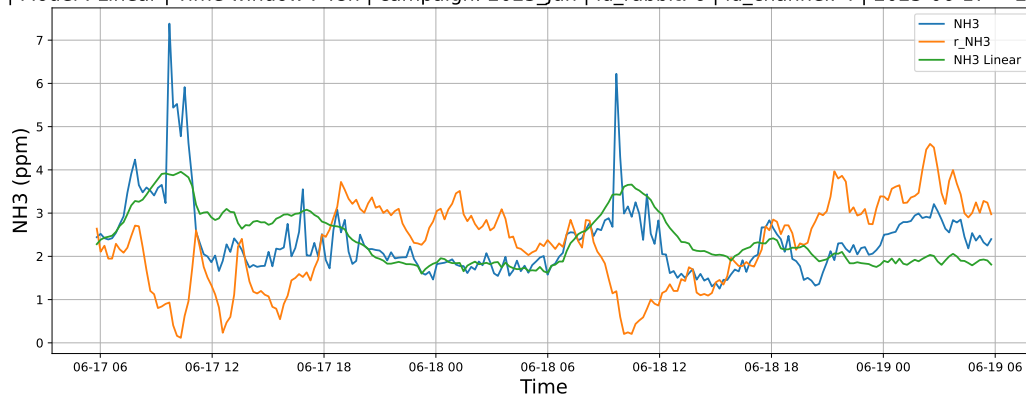


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

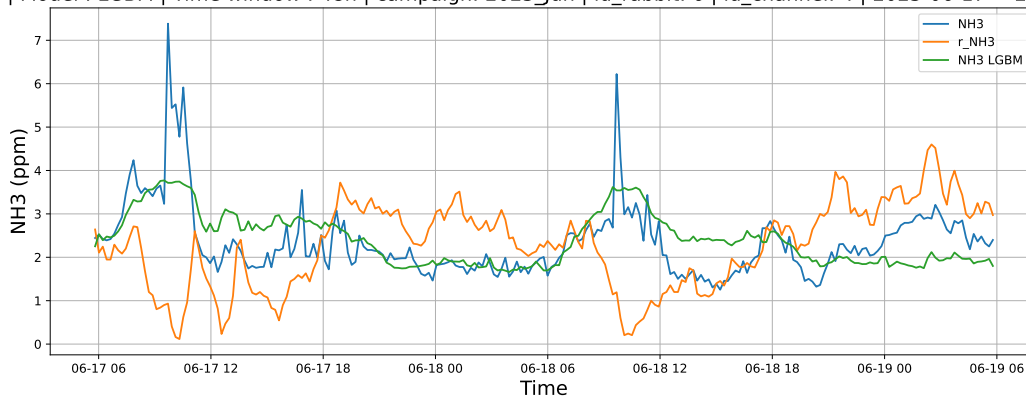


4.1.4.2 Time window : 48

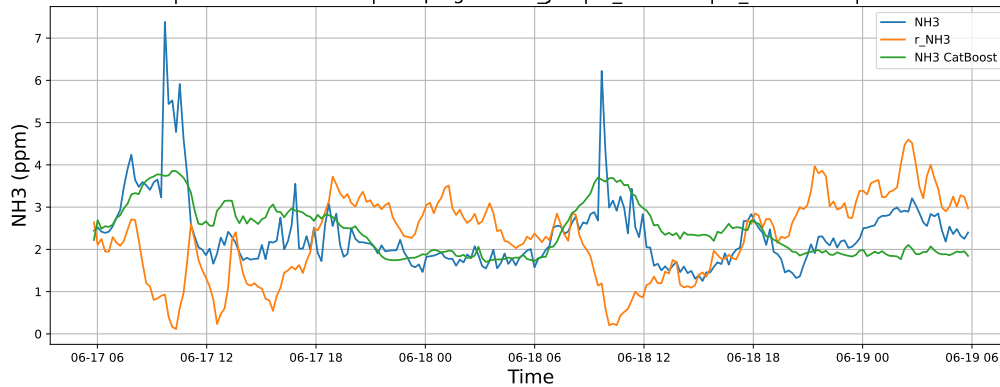
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

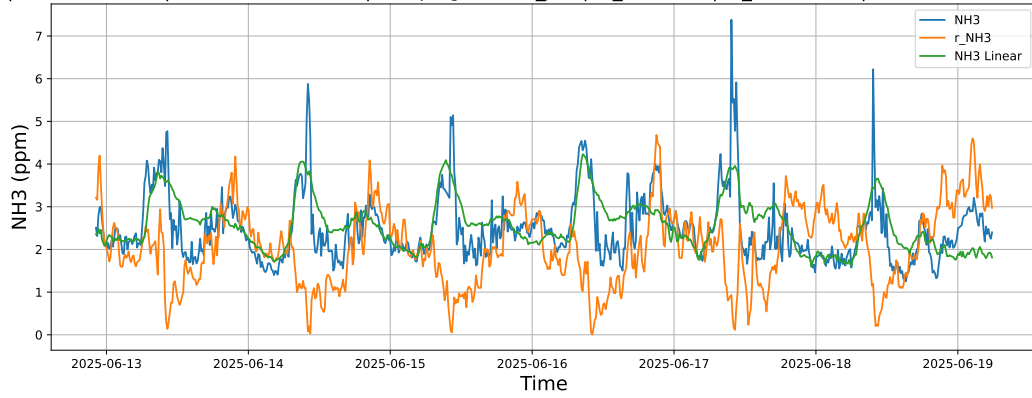


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

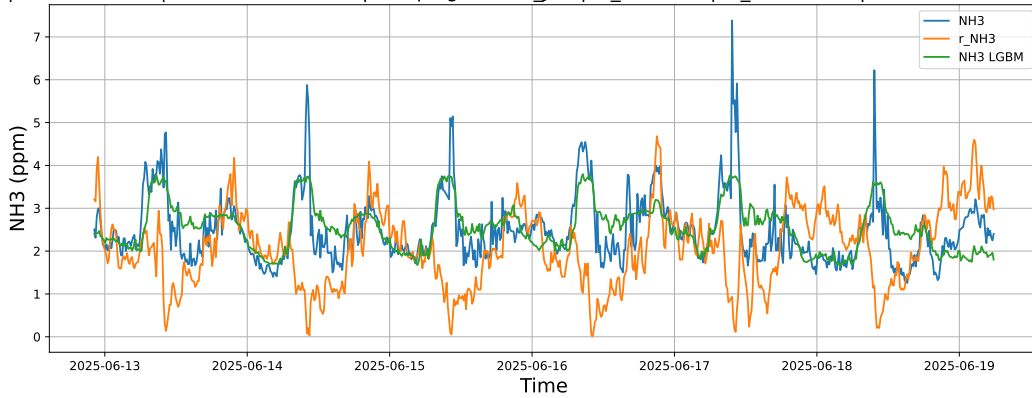


4.1.4.3 Time window : ALL

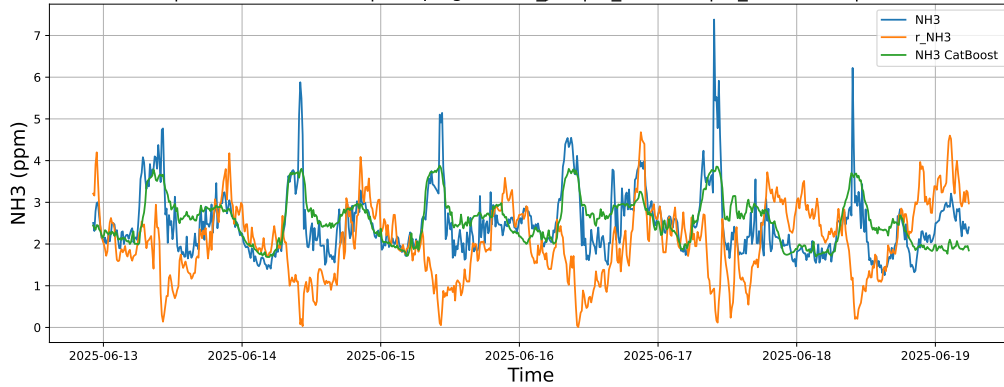
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



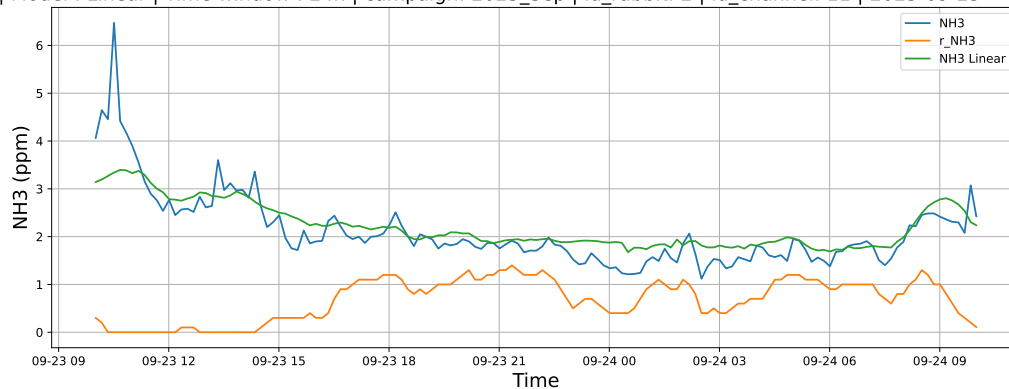
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



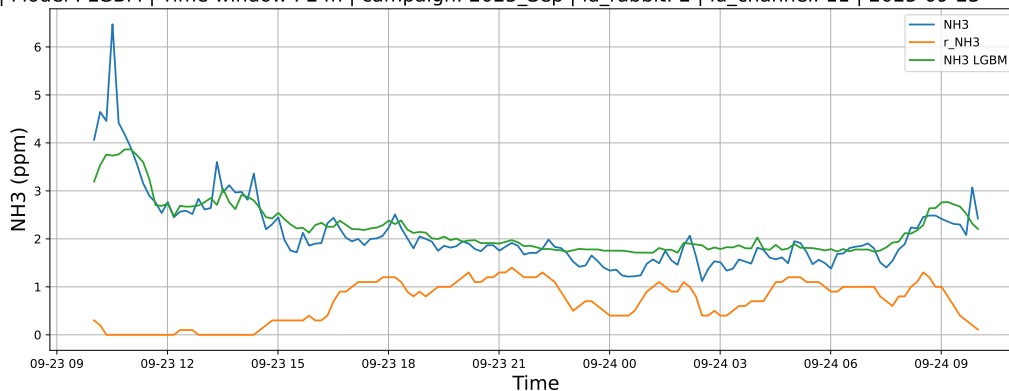
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

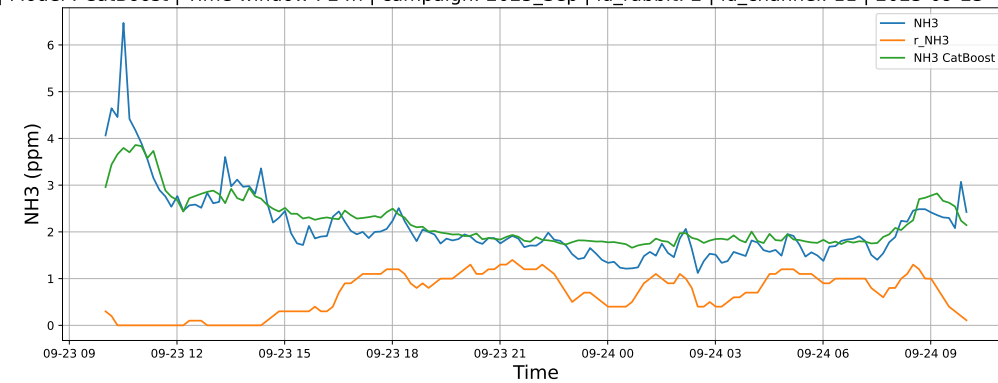
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

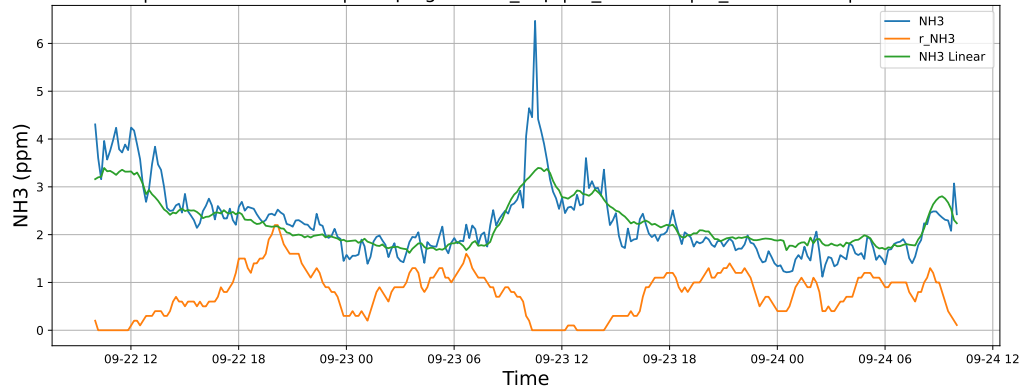


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

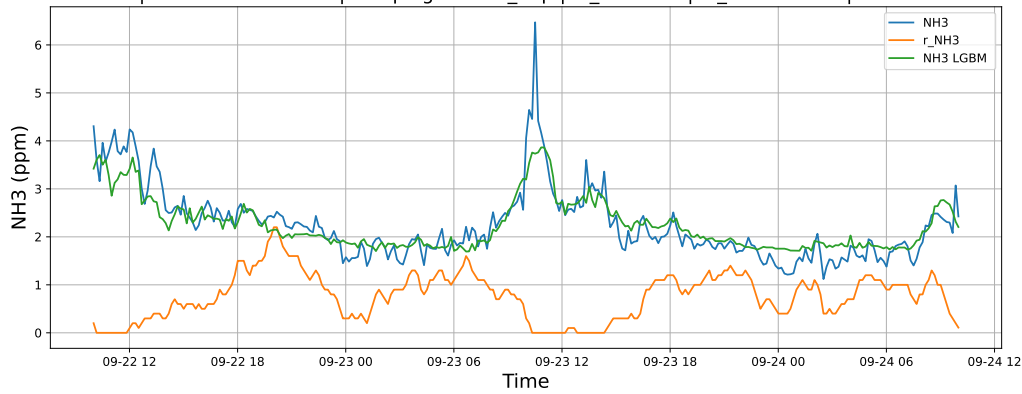


4.1.5.2 Time window : 48

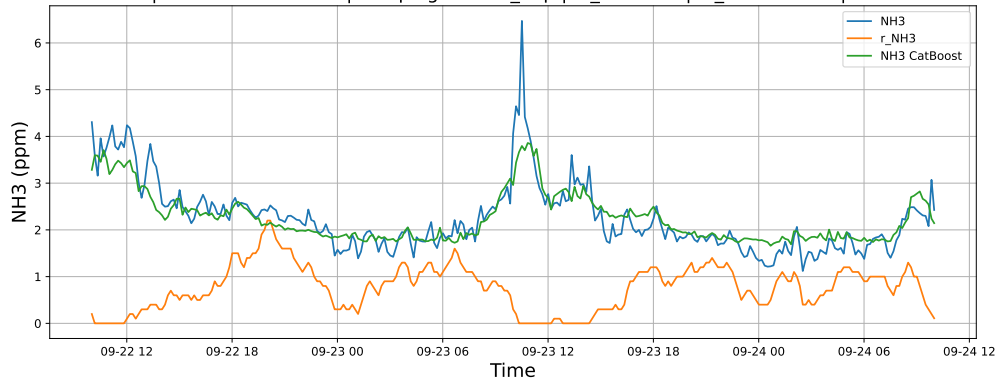
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

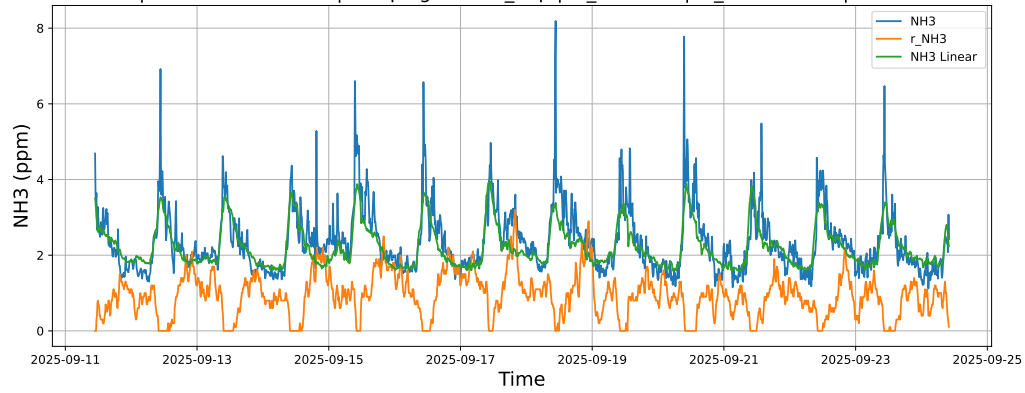


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

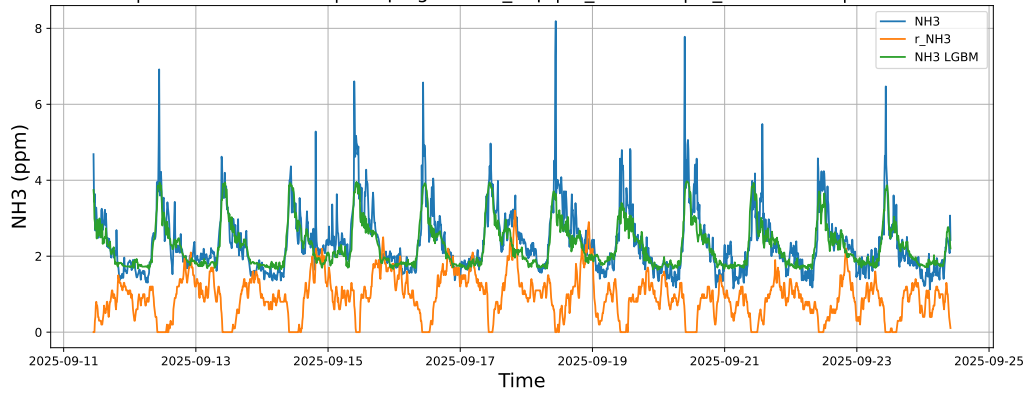


4.1.5.3 Time window : ALL

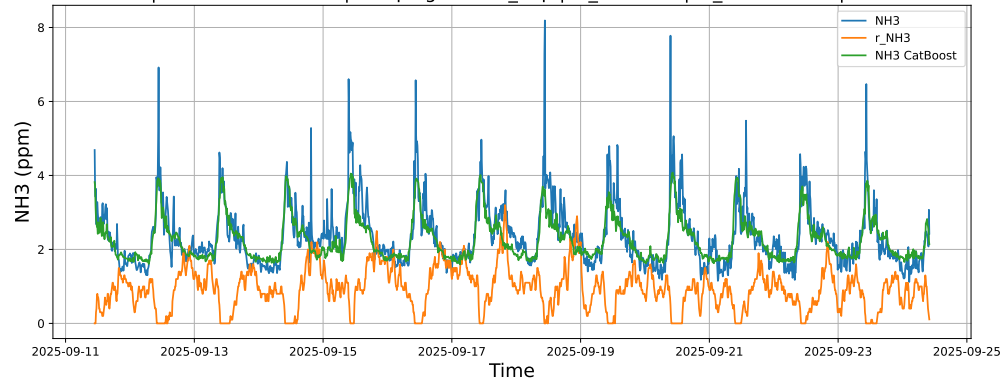
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



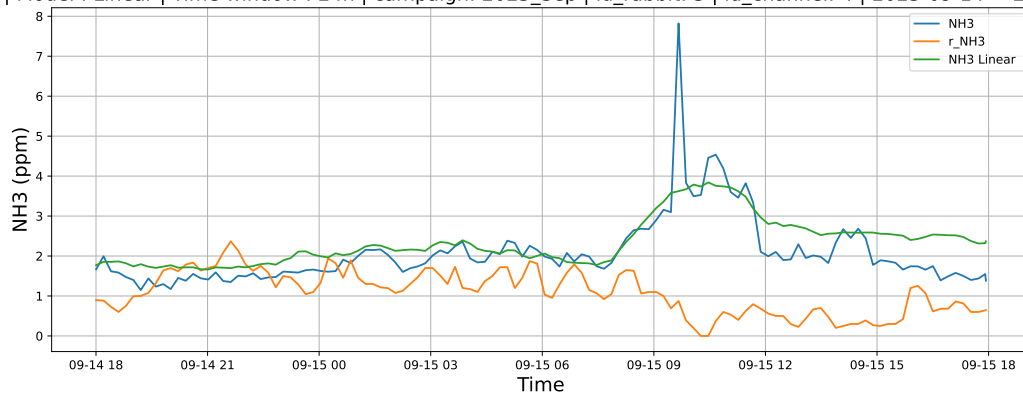
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



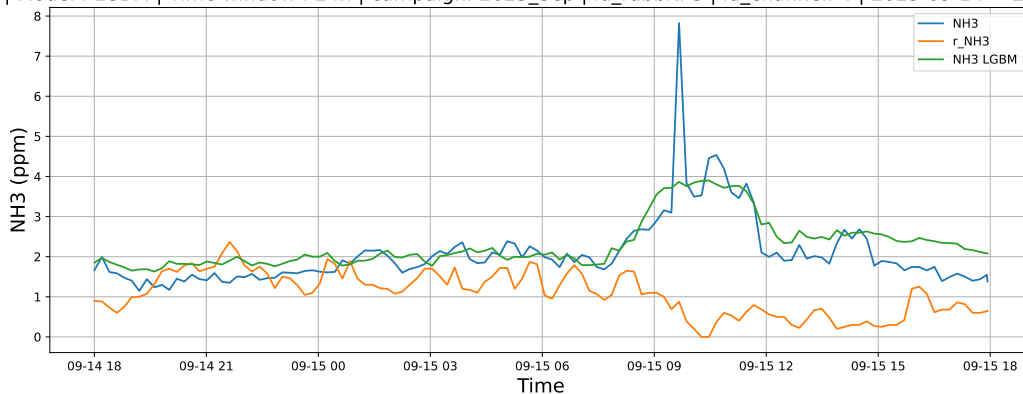
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

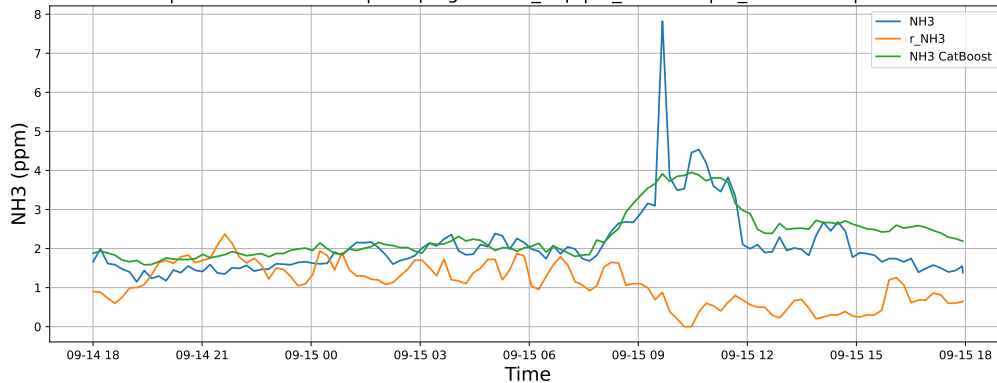
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

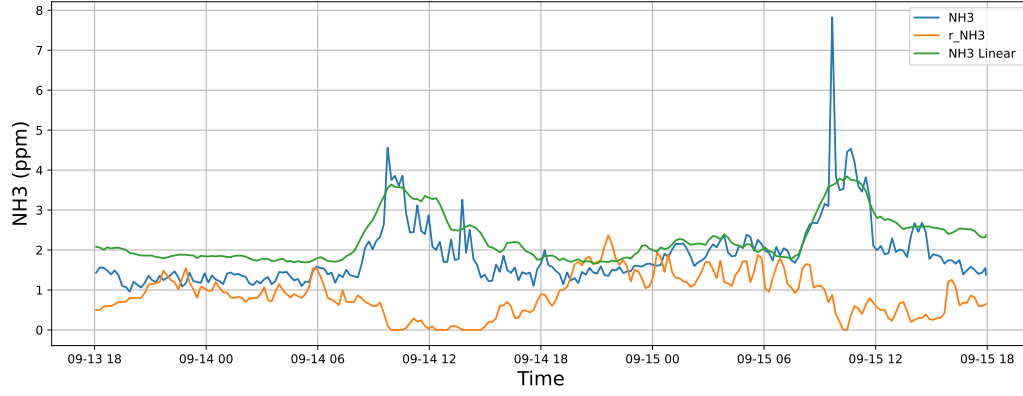


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

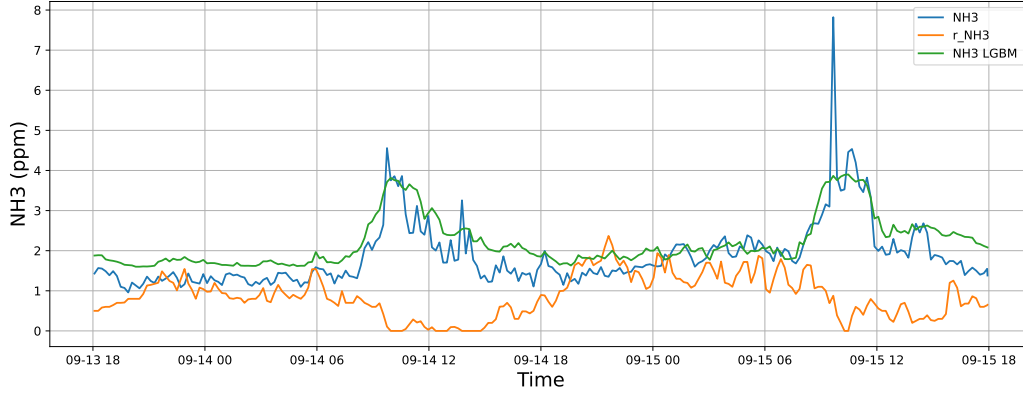


4.1.6.2 Time window : 48

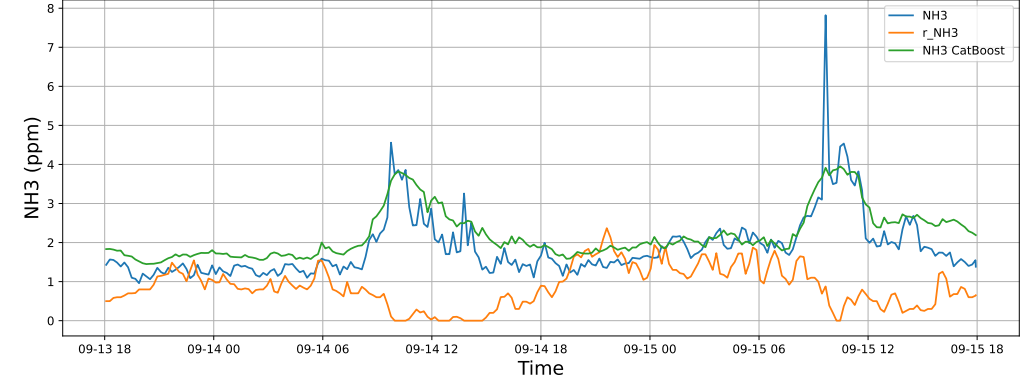
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

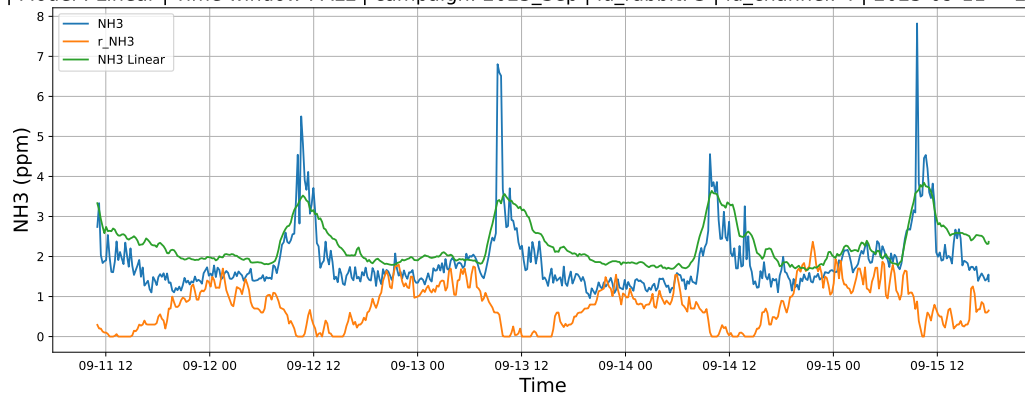


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

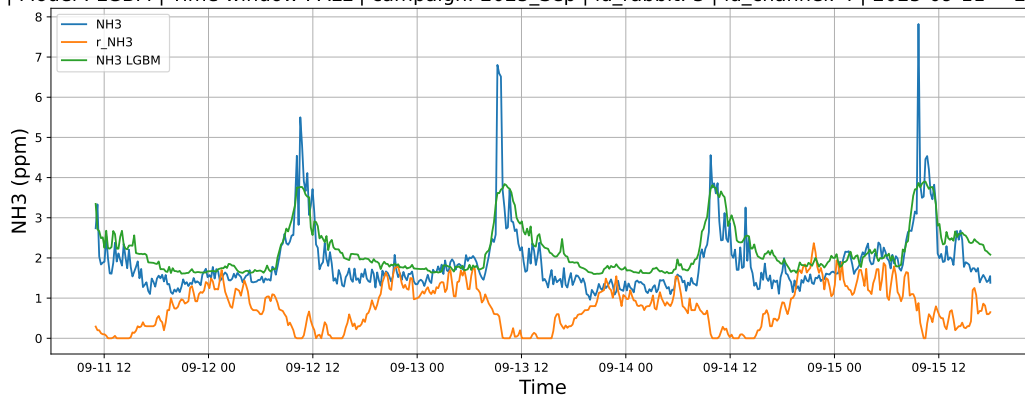


4.1.6.3 Time window : ALL

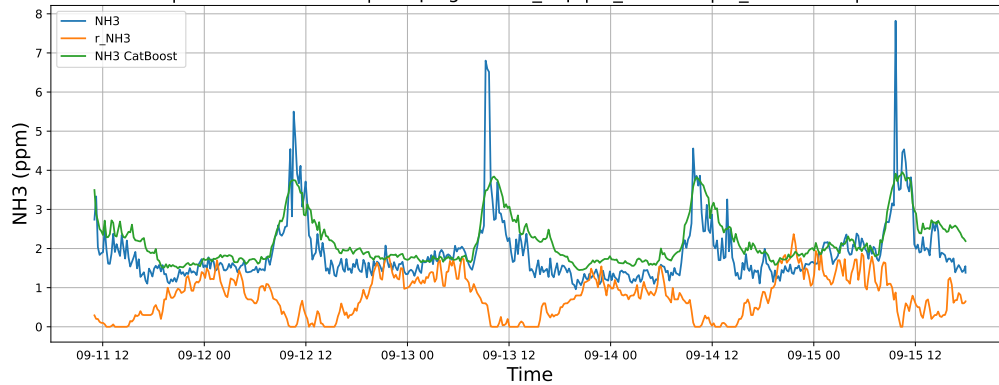
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



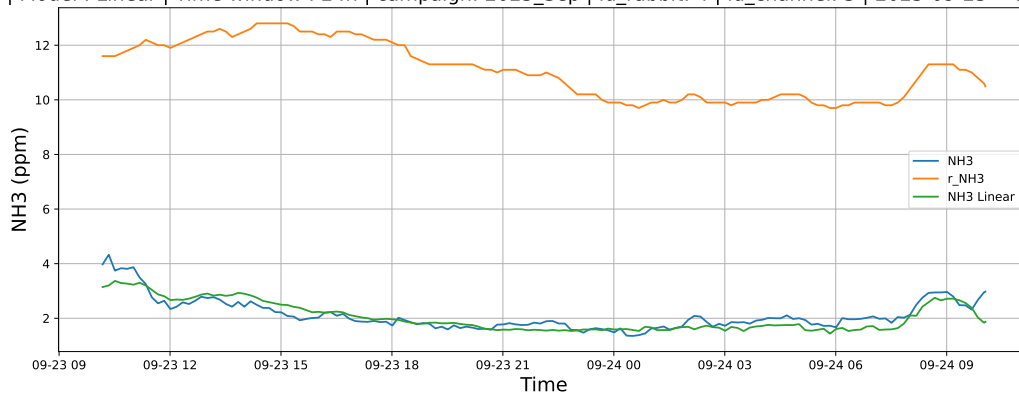
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



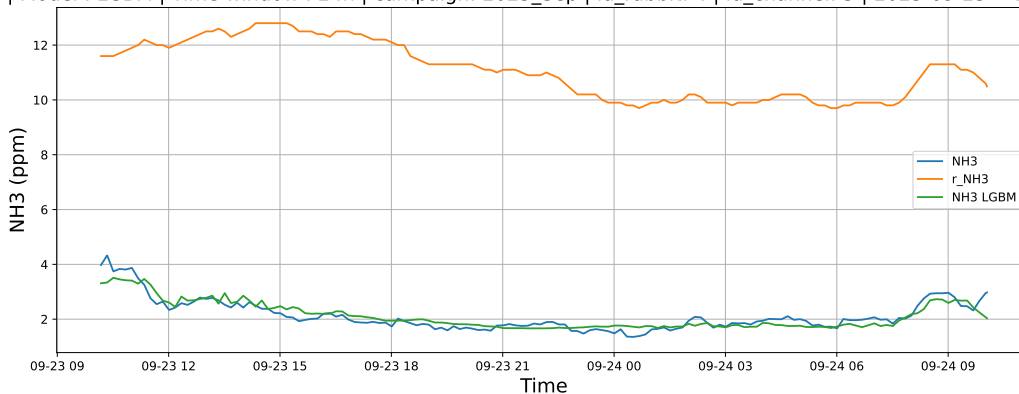
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

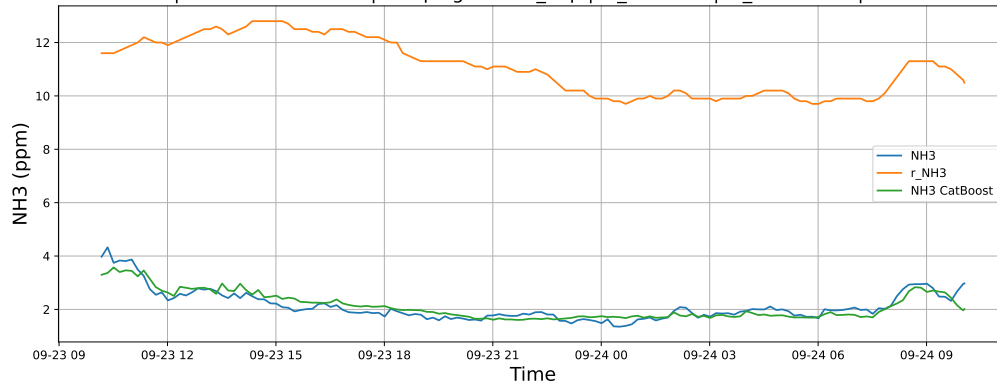
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

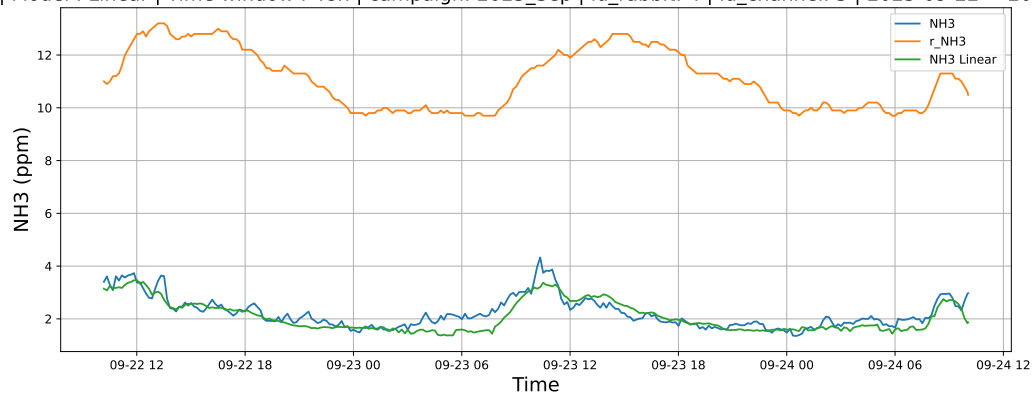


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

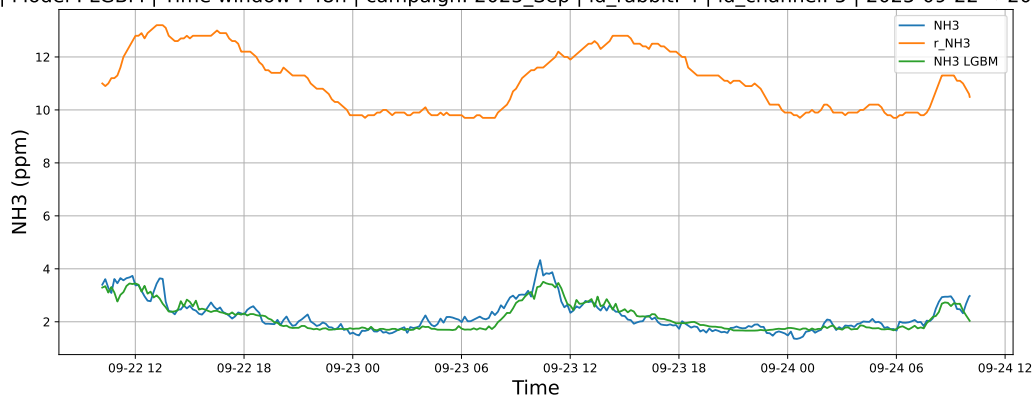


4.1.7.2 Time window : 48

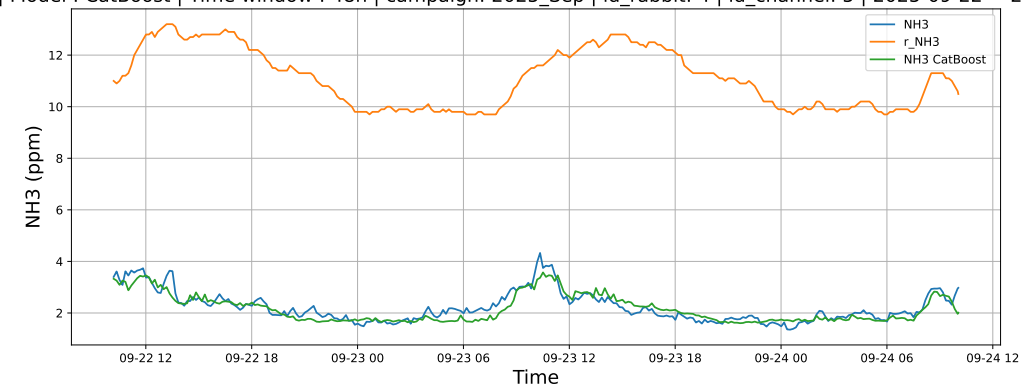
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

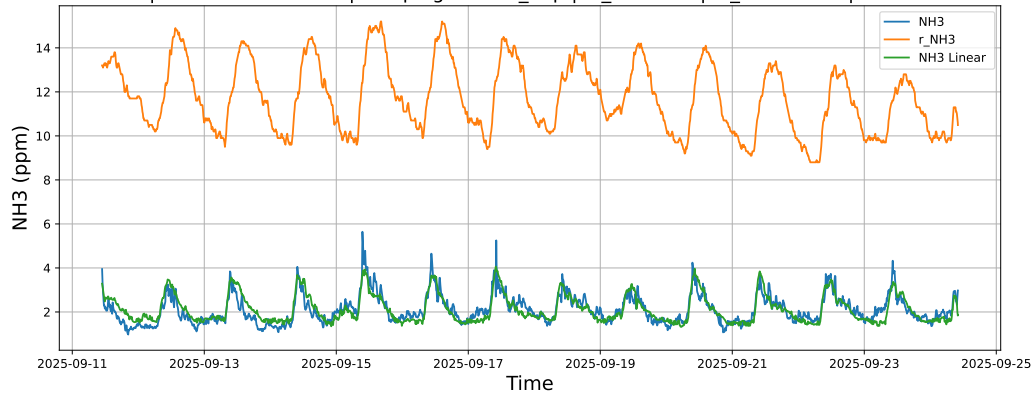


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

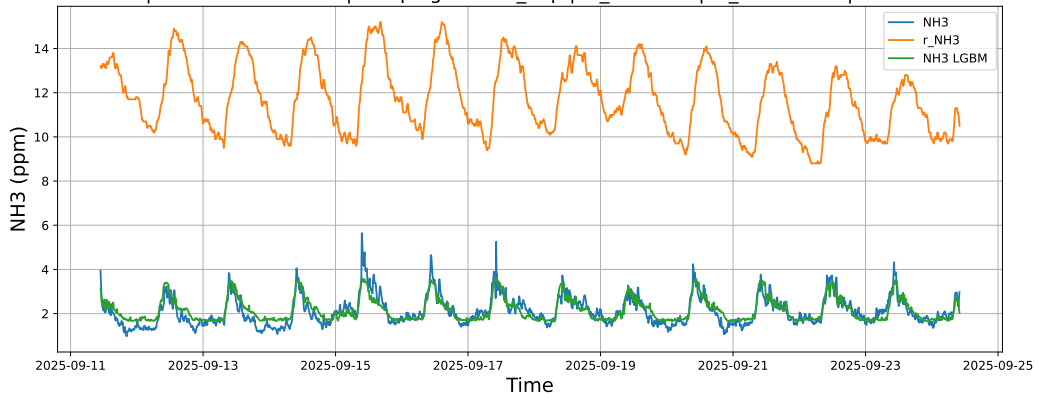


4.1.7.3 Time window : ALL

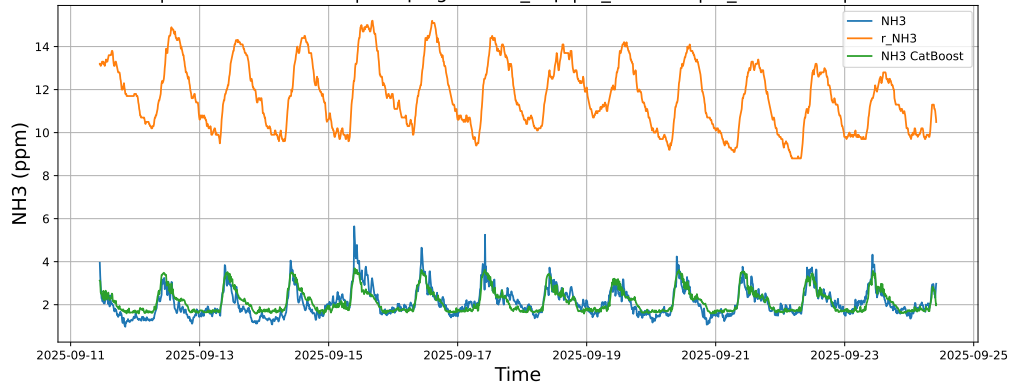
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



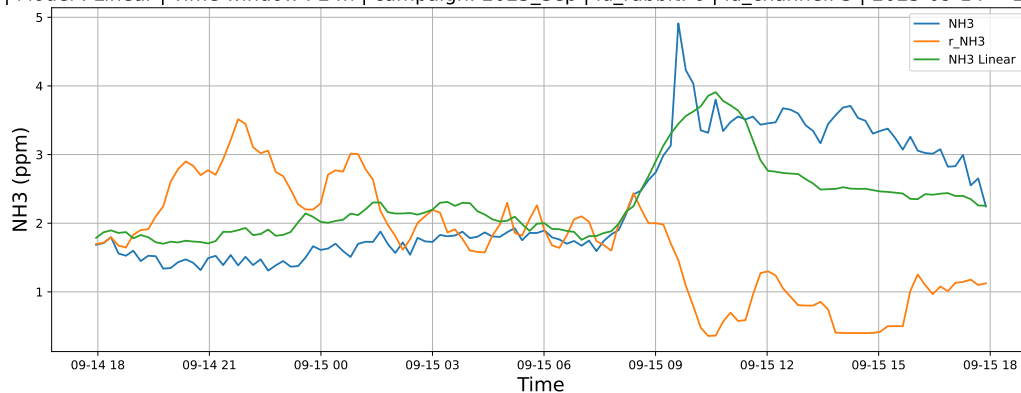
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



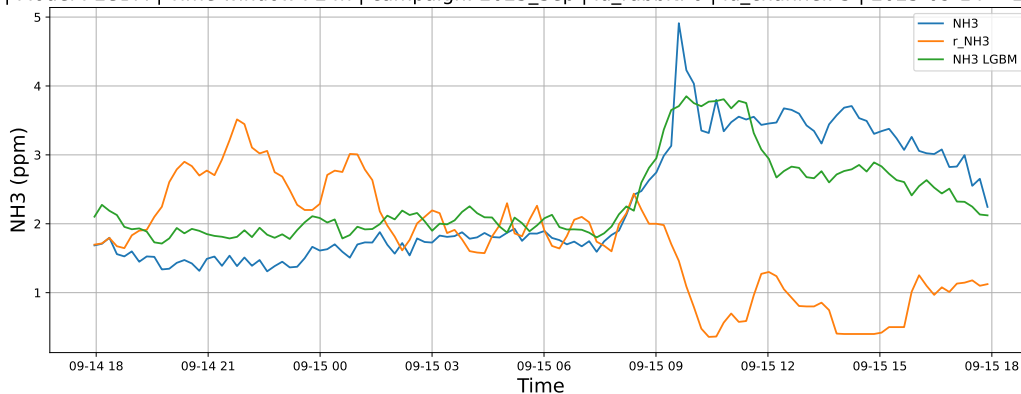
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

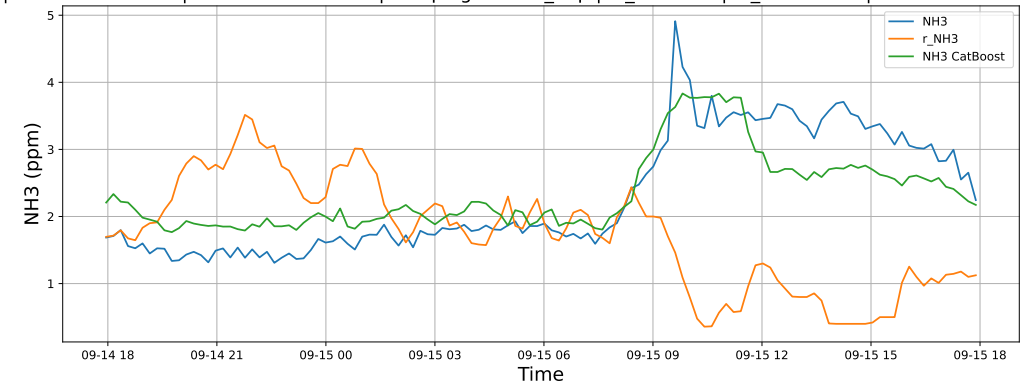
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

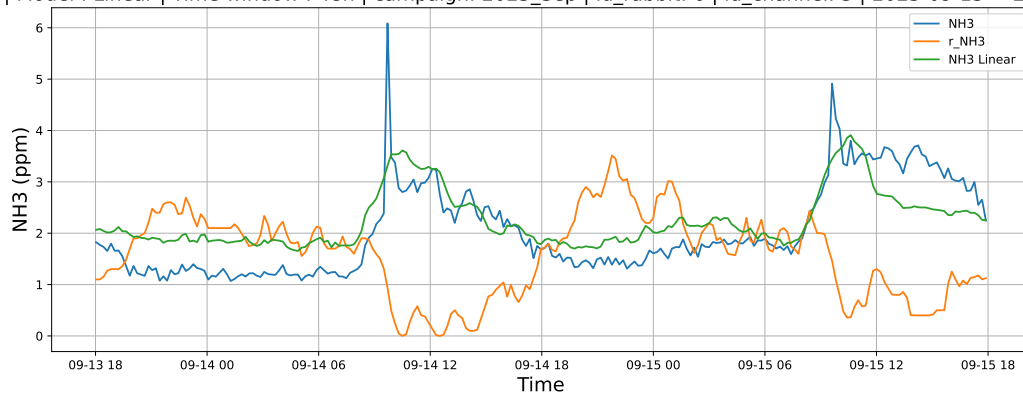


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

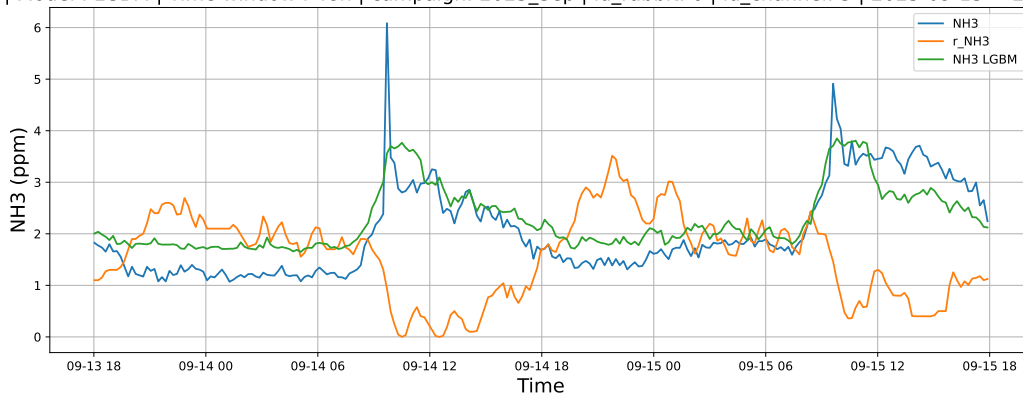


4.1.8.2 Time window : 48

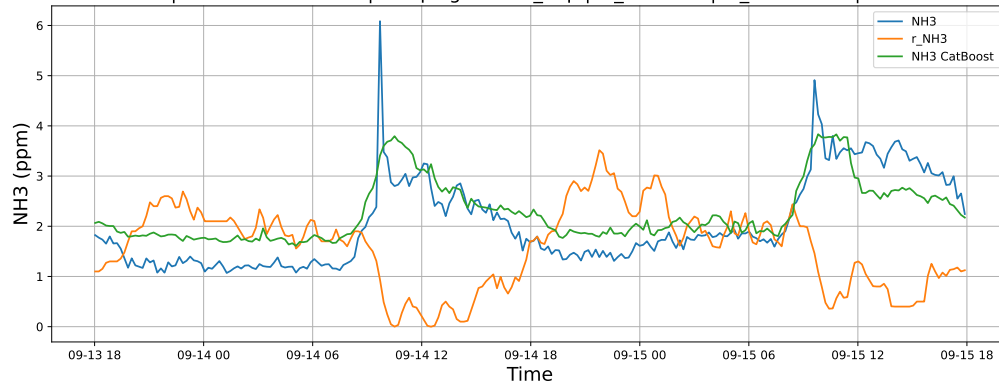
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

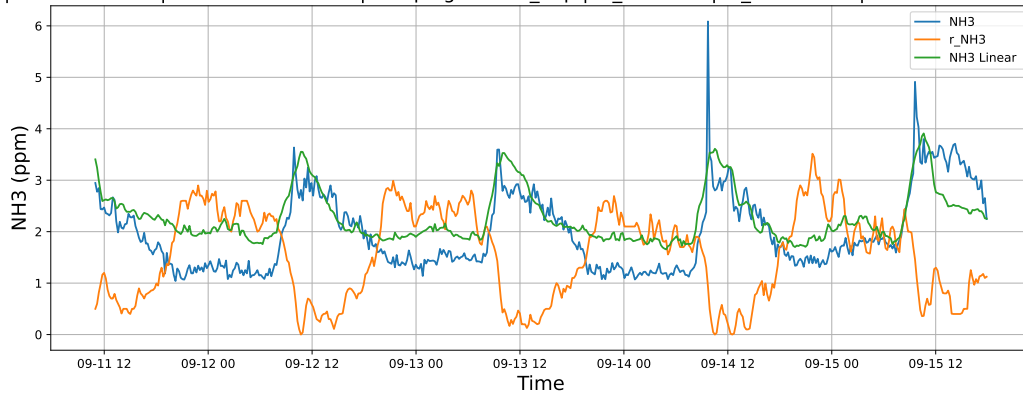


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

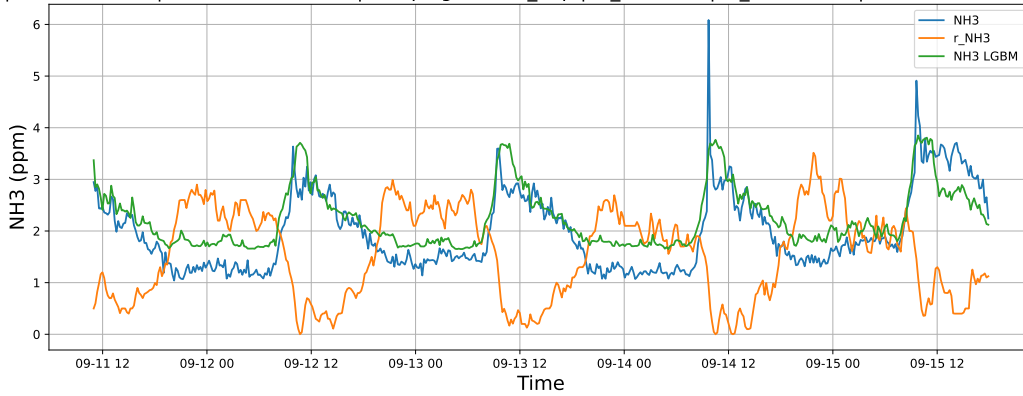


4.1.8.3 Time window : ALL

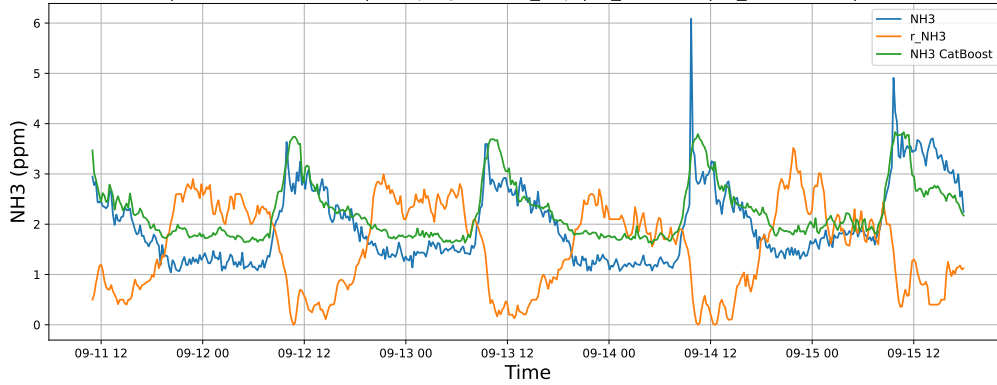
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

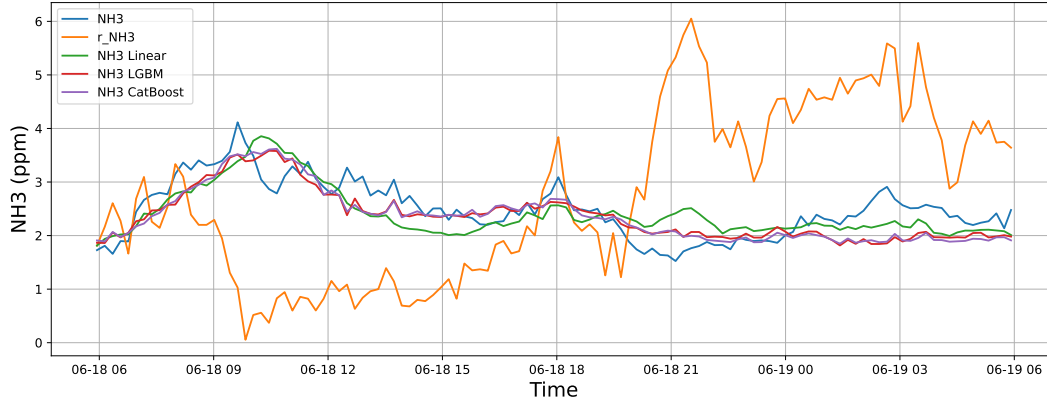


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

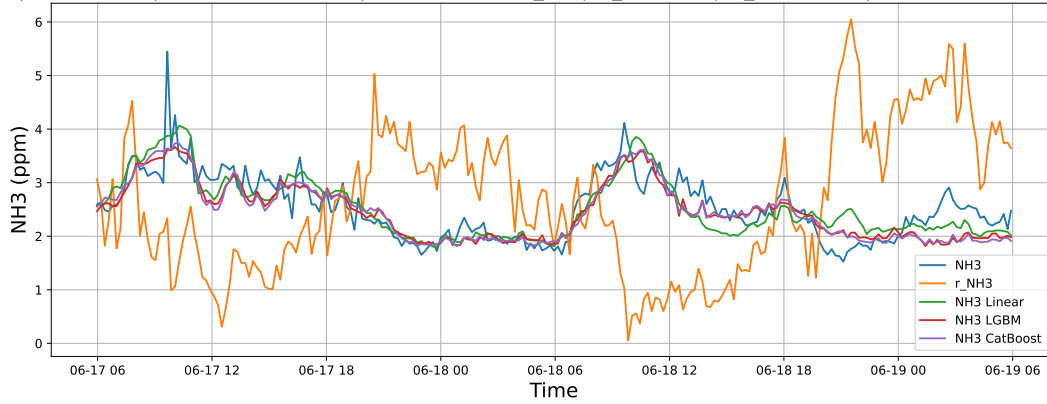
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



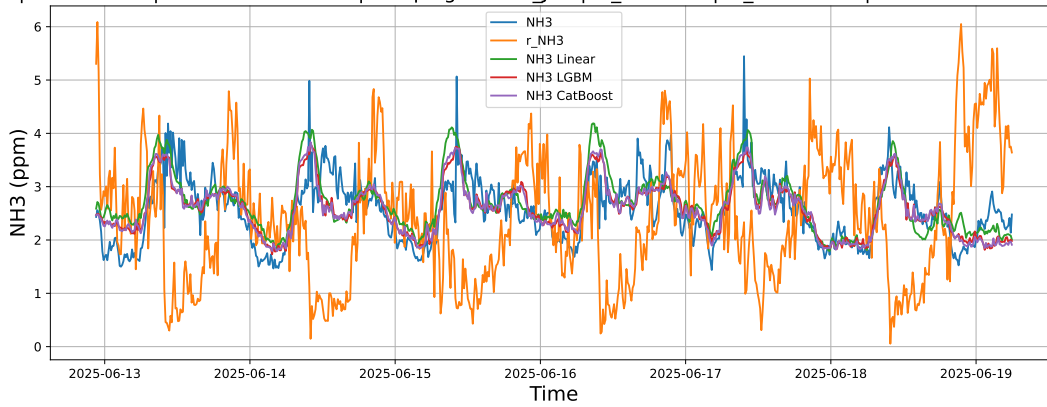
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

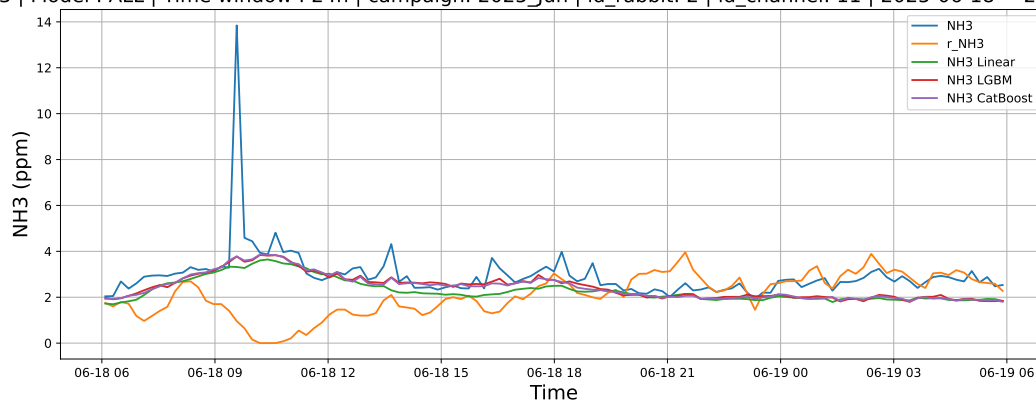
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

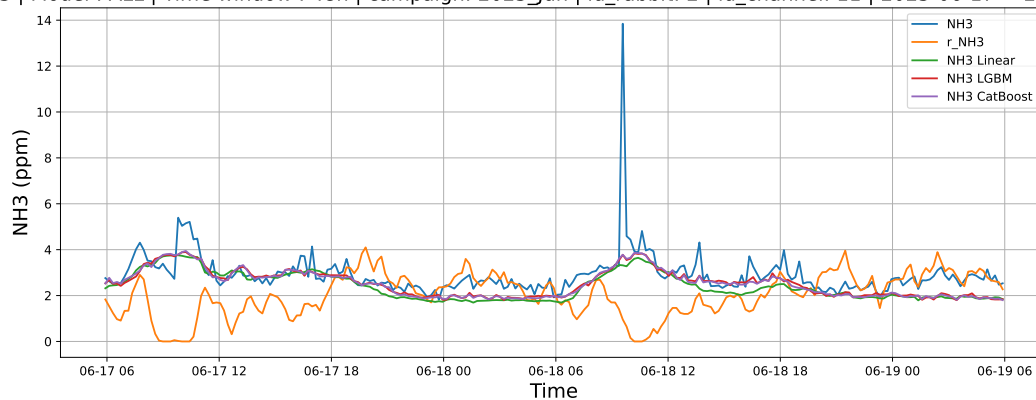
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



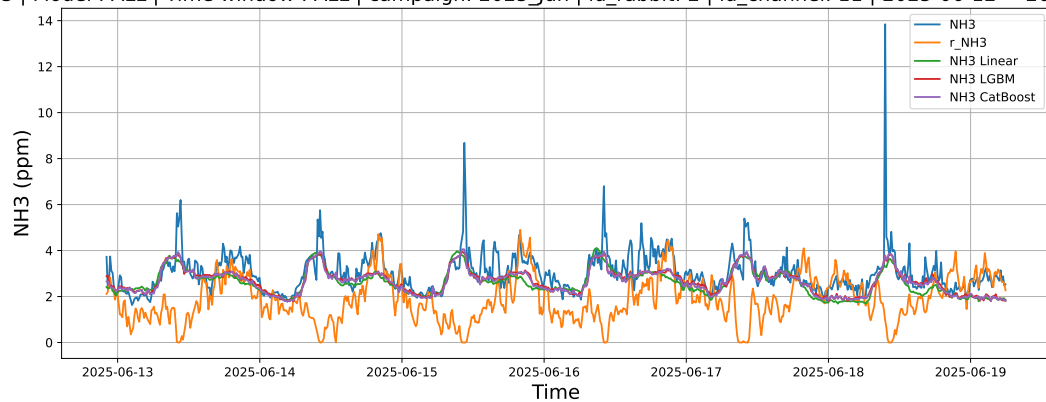
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

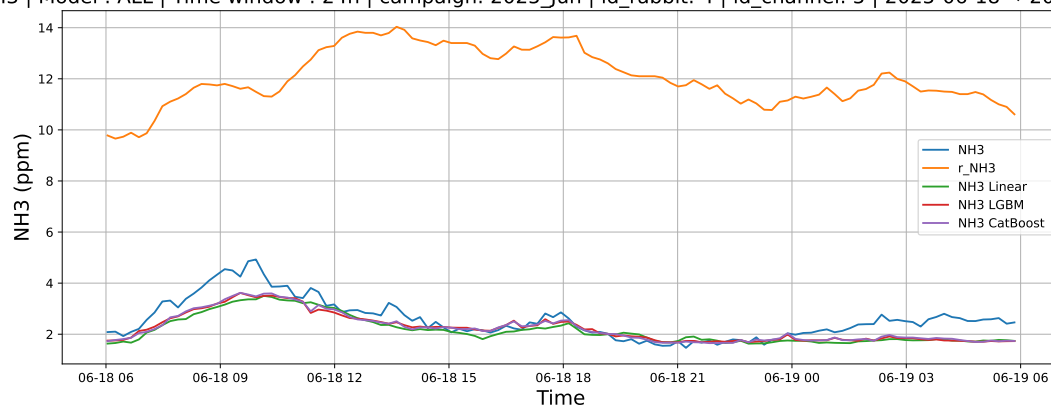
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

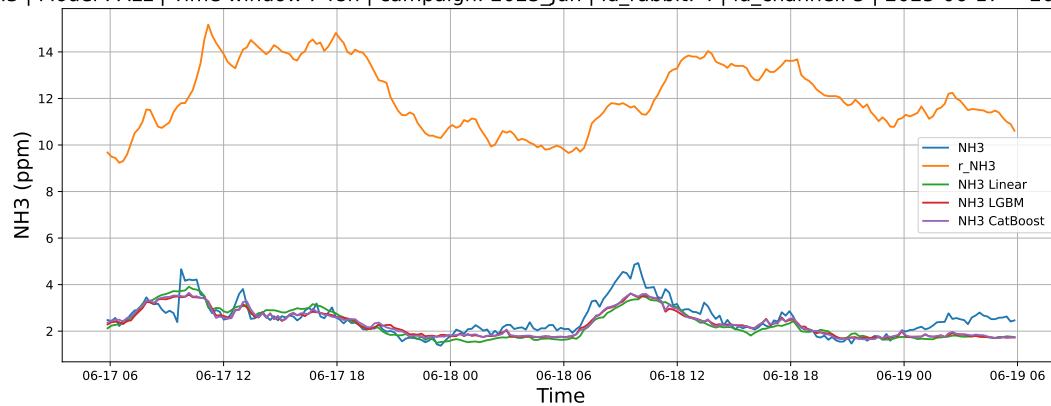
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



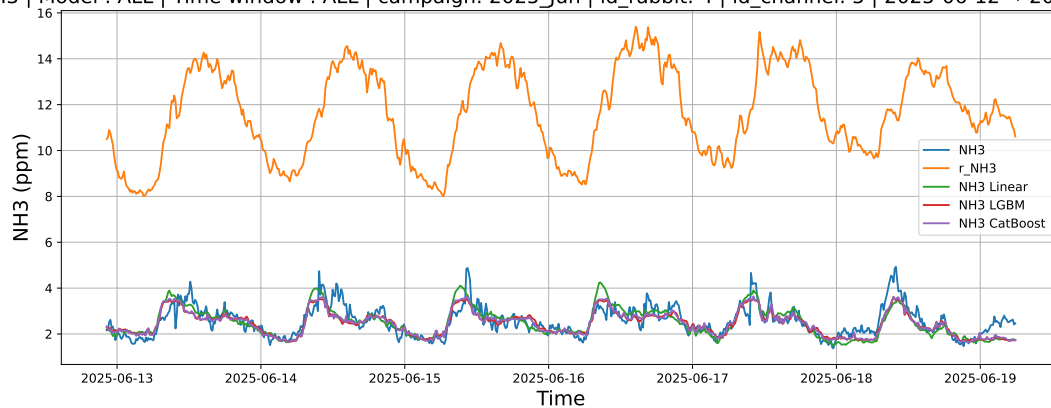
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

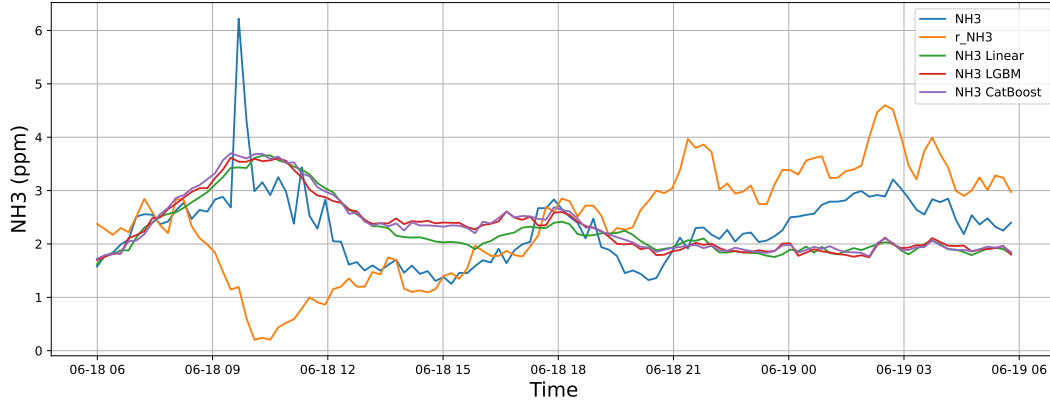
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

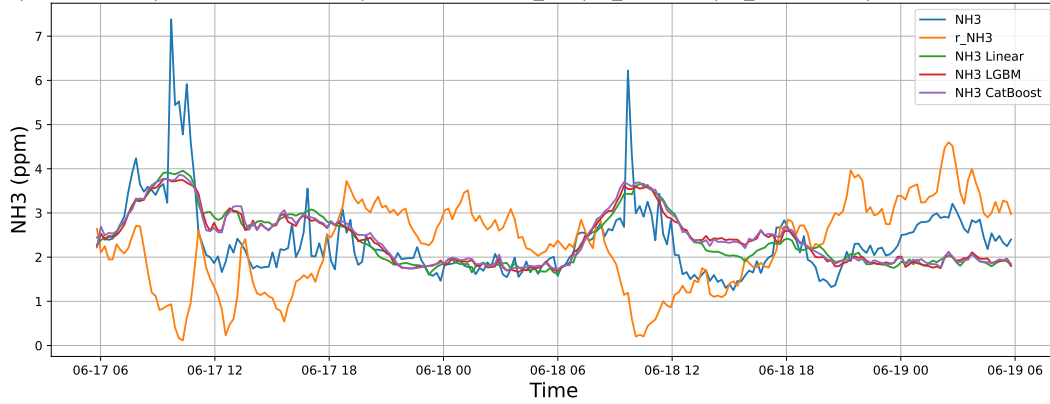
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



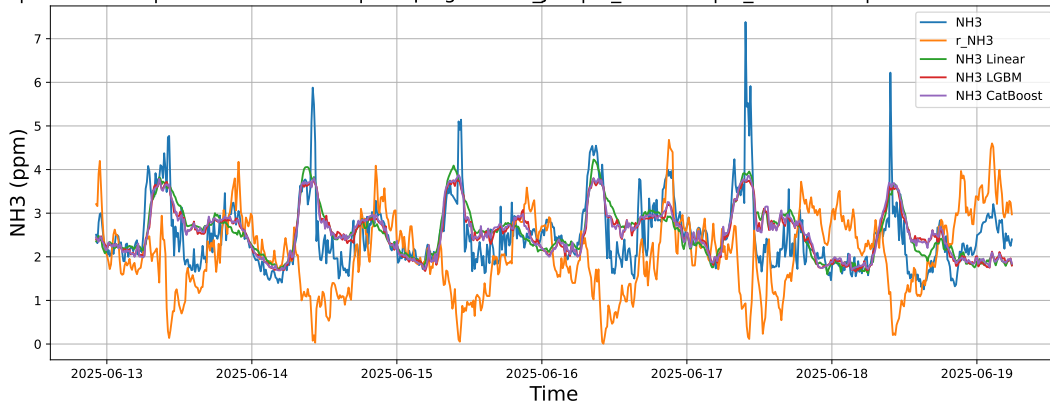
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

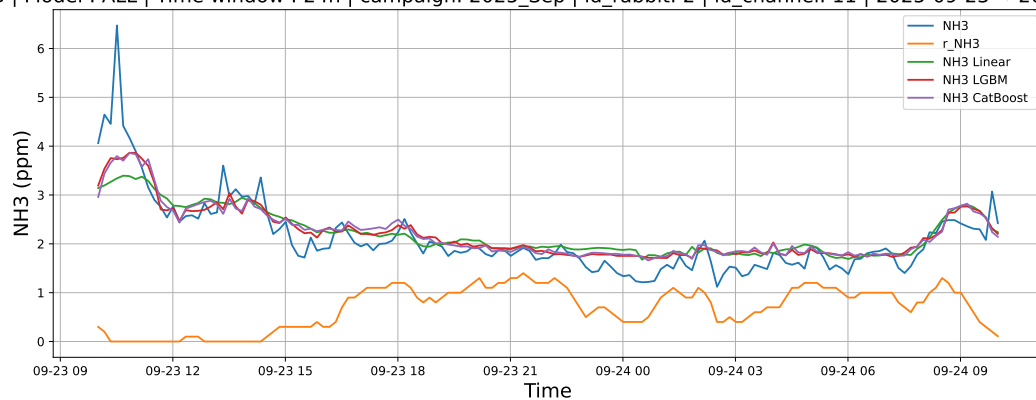
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

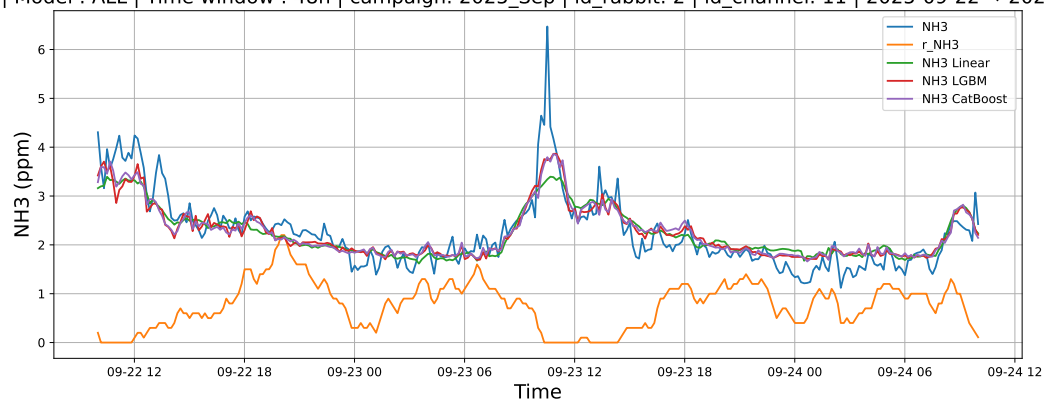
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



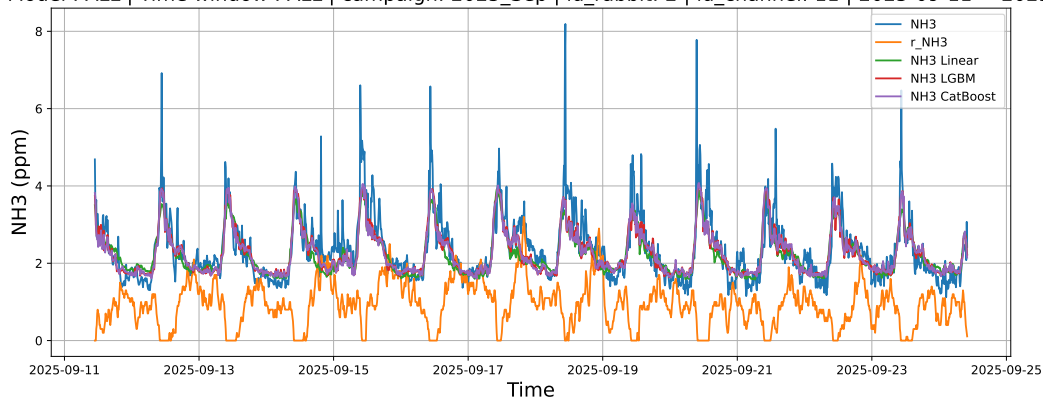
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

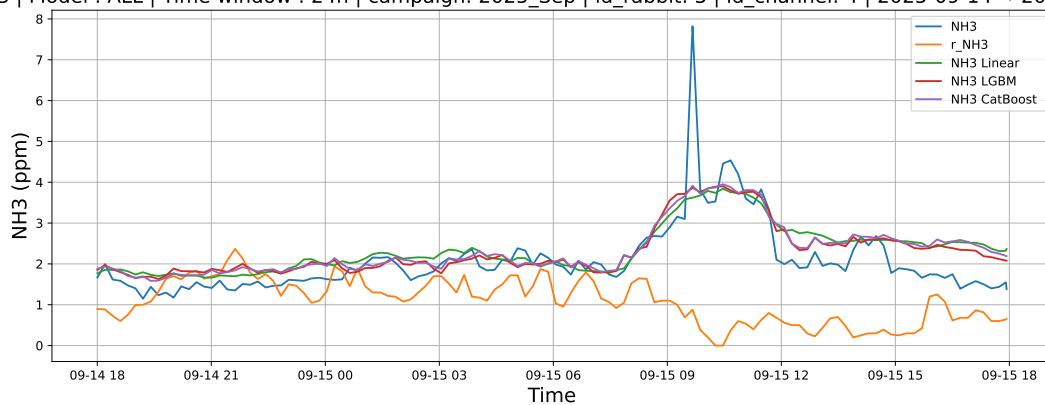
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

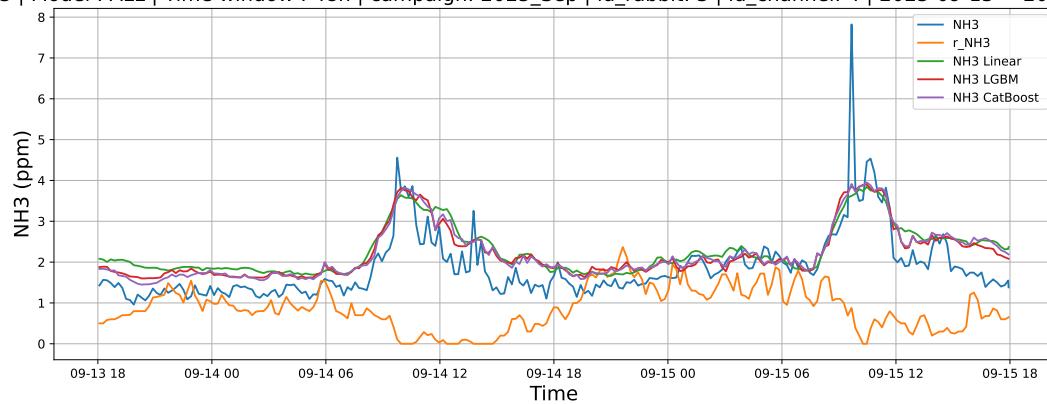
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



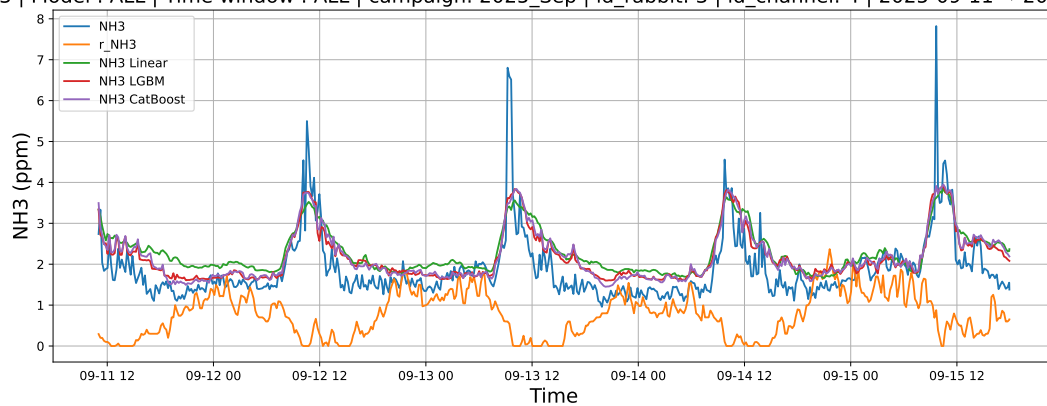
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

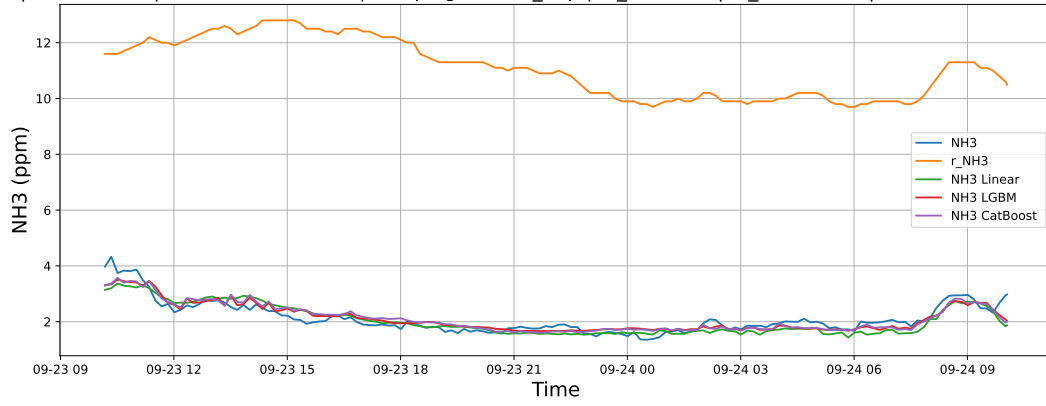
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

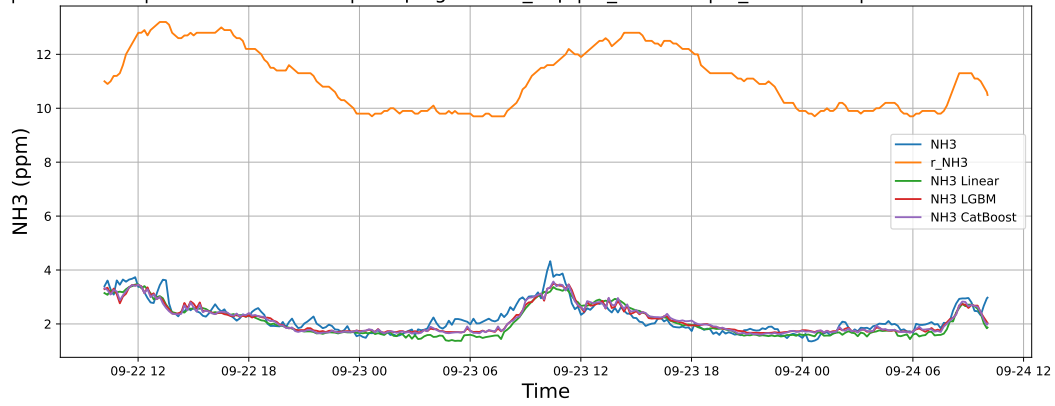
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



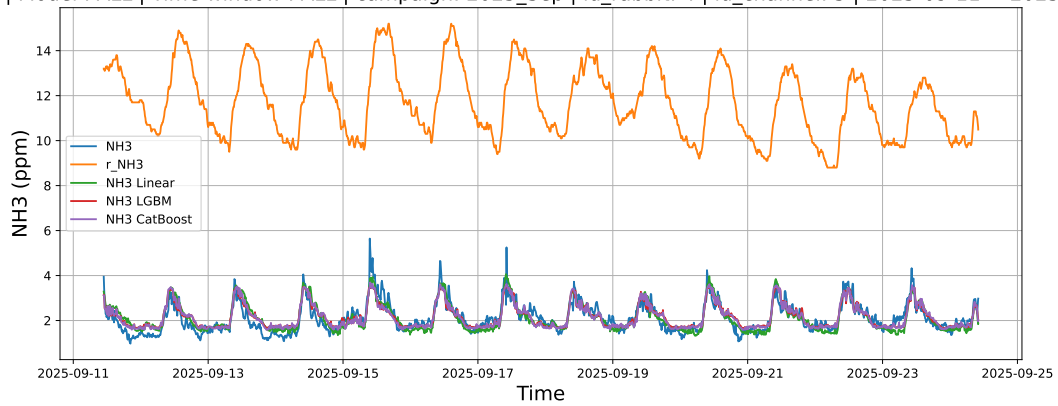
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

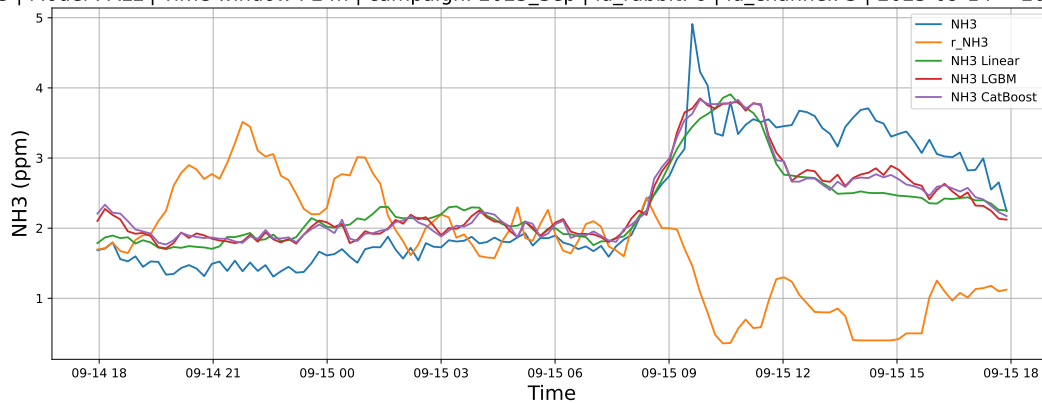
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

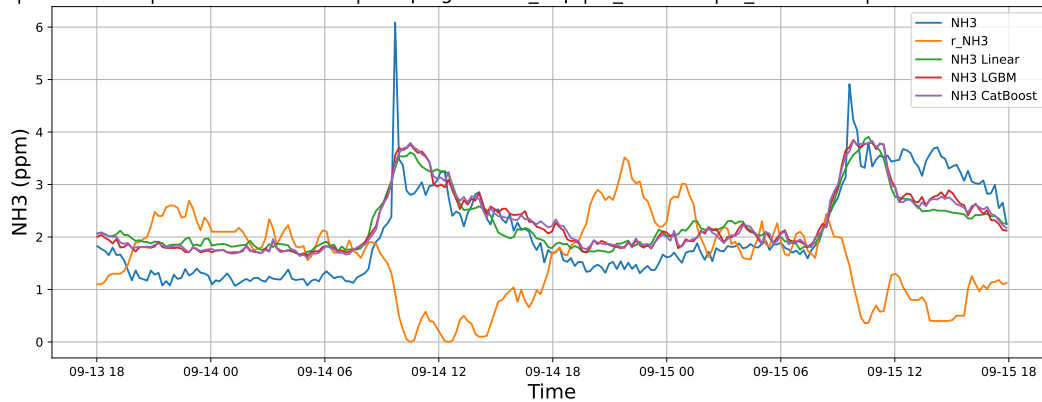
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

